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Comparative life cycle assessment of end−of−life strategies for post−consumer polylactic acid waste: Environmental trade−offs and uncertainty analysis

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ABSTRACT

The sustainable management of post−consumer polylactic acid (PC − PLA) waste is essential for reducing environmental burdens and advancing circular economy principles. This study presents a novel comparative life cycle assessment (LCA) of five end−of−life (EoL) treatment scenarios for PC − PLA waste, conducted in accordance with ISO 14040:2006 and 14044:2006 standards. The evaluated scenarios include (i) mechanical recycling, (ii) hydrolysis, (iii) methanolysis, (iv) ethanolysis, and (v) incineration. Uniquely, the study accounts for environmental credits associated with replacing virgin PLA with recycled PLA, thereby reducing reliance on both fossil−based and bio − based feedstocks. Results show that all recycling methods outperform incineration in environmental impact reduction, with mechanical recycling offering the lowest net global warming potential (GWP) at −2690 kg CO_2 eq per tonne of PC − PLA waste, owing to its low energy and chemical requirements. While mechanical recycling is energy−efficient and less chemically intensive, chemical recycling enables recovery of monomers with higher material circularity, albeit with increased toxicity and energy demand. Beyond GWP, recycling also significantly reduces agricultural land use, acidification, and eutrophication impacts by avoiding biomass cultivation. However, chemical−intensive recycling methods may introduce toxicity, warranting careful consideration. A sensitivity analysis reveals that energy use and chemical inputs are critical parameters influencing the overall environmental outcomes. Additionally, Monte Carlo uncertainty analysis highlights the variability in results stemming from methodological assumptions, data sources, and process efficiencies. This comprehensive assessment provides insights into optimizing EoL strategies for PC − PLA waste, illustrating the environmental advantages of recycling over incineration while identifying trade−offs and uncertainties to support the circular waste management.

1. Introduction

Bioplastics, as defined by the European Commission, are materials that are renewable, biodegradable, biologically produced, or a combination of these characteristics [1]. Bioplastics, derived from renewable sources such as plants, biomass, or micro−organisms, offer distinct advantages over conventional plastics by reducing reliance on limited fossil resources and lowering greenhouse gas emissions [2,3]. Currently, bioplastics account for about 0.5 % of the approximately 414 million tonnes of plastic produced annually [4]. With rising demand and the emergence of more advanced materials, applications, and products, the bioplastics market is growing rapidly. The European Commission has highlighted the vital role of bioplastics in the bioeconomy and their potential to accelerate the shift toward a circular economy [5]. Polylactic acid (PLA) is an emerging bioplastic distinguished by its favourable physical and chemical characteristics, which supports its application across diverse sectors, including packaging, disposable cutlery, drug delivery systems, and biomedical devices [6]. The global production capacity of PLA is anticipated to increase from 0.91 million tonnes in 2022 to over 2.4 million tonnes by 2027, driven by growing

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demand, positioning PLA as the bioplastic with the largest capacity [4]. The rapid growth in PLA production to meet rising demand will inevitably generate a significant amount of waste. Without proper management, large−scale accumulation of PLA in the environment can occur, contributing to plastic pollution and offsetting its benefits [7]. Thus, the increasing production of PLA must be accompanied by effective end−of−life (EoL) option to enhance the overall sustainability of bioplastics throughout their life cycle.

PLA waste can be treated through several EoL options, including composting, recycling, anaerobic digestion, incineration, and landfilling. Among these, composting is widely promoted due to its alignment with biodegradability claims. However, it requires industrial conditions, such as controlled temperature and humidity, and consumes additional energy, while failing to enable material recovery [8]. Moreover, degradation is often incomplete under ambient or natural environmental conditions [9,10]. Similarly, anaerobic digestion is constrained by PLA's inherent resistance to microbial activity, resulting in low biogas yields and limited effectiveness [11]. Landfilling, meanwhile, presents long−term environmental hazards, including leachate contamination, slow degradation, and potential microplastic formation [12]. Despite being marketed as biodegradable, PLA degrades very slowly in natural environments such as soil or marine settings, primarily due to its high molecular weight, hydrophobic backbone, and crystalline structure, which collectively hinder microbial access and enzymatic hydrolysis [13]. Unlike naturally occurring polymers, PLA requires elevated temperatures (≥58 °C), high humidity, and specific microbial consortia to initiate and sustain degradation—conditions typically found only in industrial composting facilities [14]. As a result, under ambient conditions, degradation is often incomplete and significantly prolonged, leading to the accumulation of persistent polymer fragments. These undegraded or partially degraded particles can contribute to microplastic pollution, posing risks to terrestrial and aquatic ecosystems as well as potential threats to human health through trophic transfer and bioaccumulation [15]. On the other hand, incineration allows energy recovery but contributes significantly to GHG emissions and primarily releases PLA's fixed carbon as CO_2 [16]. These limitations underscore the importance of developing more sustainable end−of−life solutions. In this context, recycling emerges as the most effective strategy. Recycling enables material recovery and keeps resources in continuous use, thereby reducing dependence on virgin feedstocks. By closing the loop of production and consumption, recycling supports a circular economy in which products are reused, remanufactured, and repurposed, ultimately minimizing environmental burdens and waste generation [14].

The life cycle sustainability of PLA depends heavily on effective recycling strategies, particularly mechanical and chemical recycling. These approaches reduce energy consumption, raw material use, and environmental burdens compared to virgin PLA production. For example, chemical recycling, PLA can be hydrolysed to yield up to 95 % lactic acid without catalysts at 160–180 °C, using four times less energy than conventional fermentation methods [17]. It can also be depolymerized into about 90 % cyclic lactide monomers within 6 h using zinc transesterification catalysts, which serve as high−value feedstocks for new PLA production [18]. Mechanical recycling, which retains PLA's polymer structure, is a low−energy, cost−effective method but is limited by polymer degradation and contamination, making it suitable mainly for clean, single−stream waste such as production scraps [19]. In contrast, chemical recycling enables recovery of constituent monomers or other valuable chemicals, offering a key advantage over composting where such resources are lost as CO_2 [20]. According to Piemonte et al. (2013), chemical depolymerization is significantly more energy−efficient than lactic acid production from virgin biomass through fermentation [21]. These benefits have driven increasing interest in chemical recycling as a viable circular economy solution for PLA, warranting detailed environmental evaluation through LCA to compare its performance against other EoL strategies and virgin PLA production.

Life cycle assessment (LCA) is a widely used tool to evaluate the environmental performance of various EoL options for bioplastic waste. It provides a comprehensive assessment of the environmental impacts and trade−offs, helping to identify the most sustainable waste management strategies and understand the ecological implications of different recovery and recycling processes [22]. While numerous research papers discuss the chemical recycling of PLA, only a limited number have conducted LCA studies to evaluate its environmental impacts comprehensively. For example, Aryan et al., (2021) investigated the LCA study of the EoL pathways for PLA waste through four scenarios: hydrolysis, methanolysis, ethanolysis, and incineration [23]. The findings revealed that chemical recycling methods offered superior environmental performance compared to incineration, highlighting their potential for more sustainable PLA waste management. In another study, Schwarz et al., (2023) demonstrated that depolymerizing advanced−-grade PLA waste led to significantly lower CO_2 emissions compared to other end−of−life options, positioning it as a more environmentally sustainable alternative [24]. However, despite the promising results of these studies, they primarily focused on the production of PLA monomers (i.e., lactic acid or lactate) or lactide as the final product and do not address the complete reproduction of PLA from these monomers within a single batch process. Furthermore, these studies frequently overlooked the incorporation of uncertainty analysis in inventory data, which is essential for capturing the potential variability in environmental impacts and improving the robustness and reliability of the results.

The main aim of this study is to address critical research gaps by assessing the recycled PLA (rPLA) production through three different chemical recycling of post−consumer PLA waste, namely hydrolysis, methanolysis and ethanolysis, comparing its energy demand and environmental impact to the production of virgin PLA from traditional feedstocks. It also presents a case study on the environmental performance of mechanical recycling and incineration, providing an in−depth analysis of different recycling options. While previous LCA studies have evaluated selected chemical recycling routes for PLA, they often exclude the complete closed−loop process and lack proper uncertainty quantification. This study uniquely addresses these gaps by modelling the full recycling pathway from post−consumer PLA to regenerated PLA, and by incorporating both sensitivity and Monte Carlo uncertainty analyses to enhance result reliability. This combined approach ensures a more comprehensive and robust comparison, offering valuable insights into identifying the most sustainable PLA waste recycling option.

2. Methodology

Life cycle assessment methodology was utilized to evaluate the environmental performance of potential recycling pathway for PLA waste, in accordance with the standards outlined in ISO 14040:2006 and ISO 14044:2006 [25,26]. The assessment followed five key phases: goal and scope definition, life cycle inventory analysis, impact assessment, uncertainty analysis, and interpretation. The impact assessment was carried out by employing the ReCiPe 2016 v1.03 Midpoint (Hierarchist) method, a widely adopted approach for environmental impact evaluation [27], implemented within the OpenLCA 2.2.0 software.

2.1. Goal and scope definition

The primary objective of this study was to evaluate the environmental impacts associated with various recycling technologies for post−consumer polylactic acid (PC − PLA) waste, benchmarking them against conventional thermal treatment in a municipal solid waste incineration (MSWI) facility. The study focused on recycling techniques, such as mechanical recycling, hydrolysis, methanolysis and ethanolysis, which were assessed in comparison to typical thermal treatment. A detailed description of these scenarios and the flows and providers used in this study are provided in Table S1 and Table S2 respectively. The three solvent−based recycling technologies were at comparable technology readiness levels (TRL 2–4), indicating their early−stage

development status. Despite this similarity in technological maturity, the recyclates produced through these processes varied significantly in terms of material properties and quality. Furthermore, each recycling technology had distinct feedstock requirements, particularly regarding the purity of the input PC – PLA waste, which complicated a direct performance comparison across technologies. Nevertheless, all the recycling scenarios utilized PC – PLA waste as their primary input, enabling a comparative analysis of the potential environmental impacts associated with their respective treatments.

In addition to evaluating the environmental performance of these technologies, the study also sought to identify critical environmental hotspots that offer opportunities for future optimization. A key aspect of this analysis was the variation in impurity tolerance among the different recycling processes, which played a crucial role in determining both

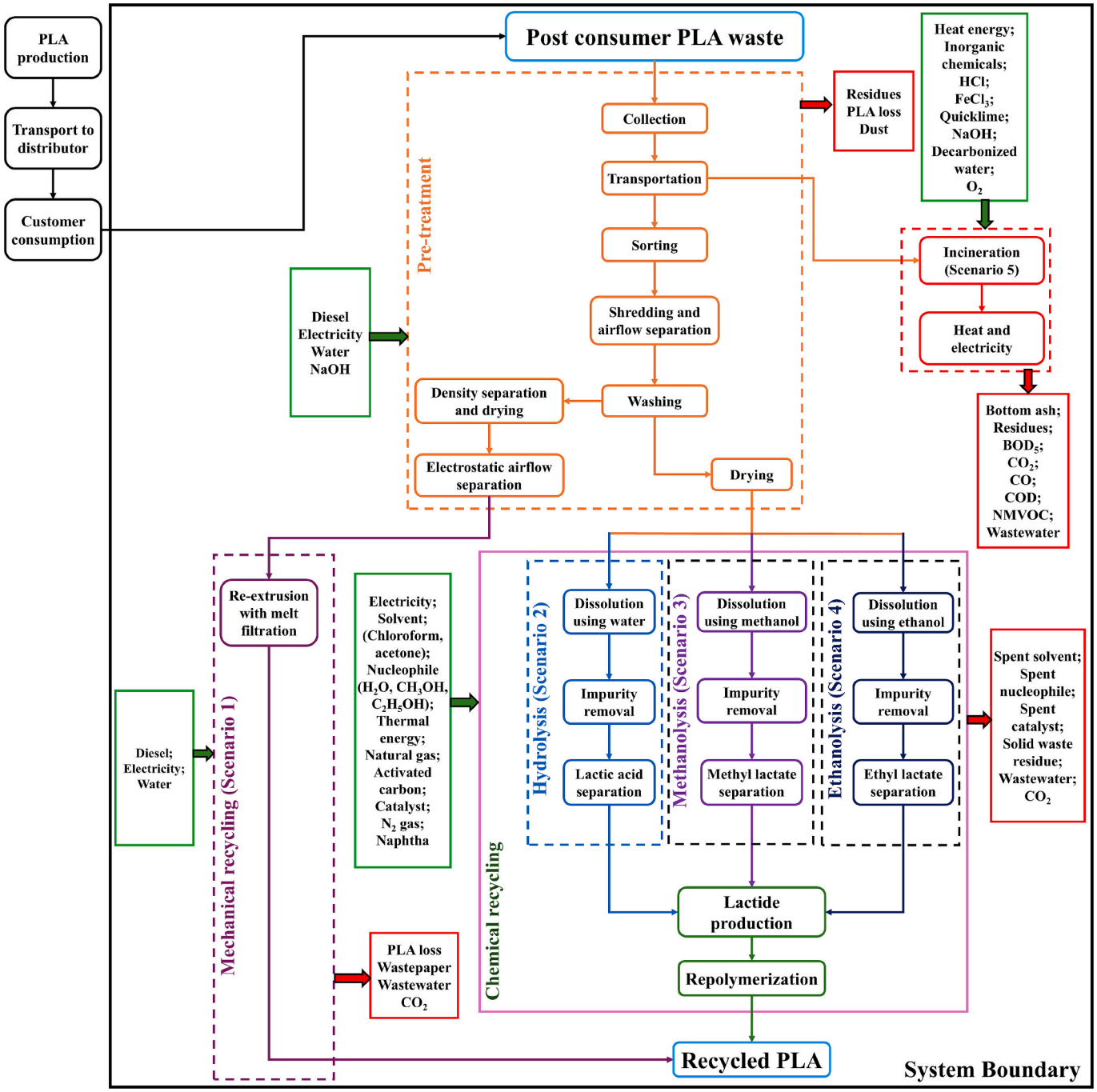


Fig. 1. System boundary of post–consumer PLA waste management pathways, including pre – treatment, mechanical recycling (Scenario 1), chemical recycling via hydrolysis (Scenario 2), methanolysis (Scenario 3), ethanolysis (Scenario 4), and incineration (Scenario 5). All pathways include a common pre – treatment stage followed by scenario–specific recycling or disposal processes. Processes are enclosed in color–coded boundaries to indicate their functional grouping.
Note: The figure emphasizes downstream end–of–life processes; upstream stages such as PLA production, transportation to distributors, and consumer use are excluded from the system boundary in accordance with study objectives. Inputs (e.g., diesel, electricity, water, chemicals) are highlighted in green boxes and arrows, and outputs (e.g., rPLA, CO_2 emissions, wastewater, solid residues) are shown in red boxes and arrows. Output flows represent recovered materials, by–products, or environmental emissions. Emissions are shown with outward arrows, and final recovered products or residues are directed to their respective fates. Key life cycle inventory data for each scenario are detailed in Table 1.

their technical feasibility and environmental impact. The outcomes of this study are expected to inform future improvements in the design and operation of PLA recycling technologies, contributing to the development of more resource−efficient and sustainable waste management strategies.

2.1.1. Functional unit

The functional unit (FU) was defined as the treatment of 1 t of PC − PLA waste to compare PLA recycling with its thermal treatment in an MSWI plant. Given the potential variation in the proportion of PLA across various waste streams, the FU was defined based solely on the PLA content, with all other materials in the waste stream excluded from the analysis.

2.1.2. System boundaries

The LCA study employed an attributional approach, concentrating on the EoL stage of PC − PLA waste. It evaluated the environmental impacts associated with (i) the collection, sorting, and transportation, (ii) pre − treatment processes, (iii) specific EoL treatment methods, and (iv) the production of recyclates as outputs from recycling processes. Four recycling scenarios (Scenarios 1–4) were examined to assess the environmental performance of PLA waste management, followed by a fifth scenario representing thermal treatment. The system boundary for LCA study of PC − PLA waste is illustrated in Fig. 1. In each recycling scenario, PLA − based disposable items—such as drinking cups, food plates, and lightweight packaging (LWP) waste—were collected from domestic areas and delivered to a sorting facility, where near−infrared (NIR) technology was used to separate the PLA − containing fraction. This fraction was then shredded and further purified through airflow separation, washing, and drying. Additional purification steps, such as density separation and electrostatic air flotation, helped in reducing contamination from other materials (e.g., plastics, paper, and cardboard) from 25 to 30 % to below 5 %. The purified PLA waste underwent mechanical recycling (Scenario 1), where it was processed through melt filtration during extrusion. This step effectively removed up to 5 % of solid impurities, yielding high−quality recyclates.

In Scenarios 2–4, the waste feedstock undergoes a selective dissolution process to extract PLA from the stream. Materials that could not be separated by density were fed into a boiler containing a dissolution mixture formulated to eliminate contaminants such as paint, adhesives, and paper. These impurities were subsequently filtered out and subjected to appropriate treatment [28]. The dissolved PLA was then subjected to thermal treatment to remove the dissolution mixture. The solvent was continuously recycled through filtration and purification processes, enabling its reuse in subsequent cycles. Following pre − treatment, the PLA waste underwent solvolysis to generate valuable recyclates. The system boundary was extended to encompass the utilization of these products in the production of rPLA.

In Scenario 5, the collected PLA − containing waste was directly transported to MSWI plant. The energy recovery from incineration was considered, with the generated electricity and heat assumed to offset equivalent fossil−based energy sources, based on the average Australian energy mix.

2.2. Data collection and quality

Literatures were the main source for obtaining inventory data for scenario 1, focusing on relevant processes and performance metrics [16], while data for Scenario 5 was obtained from the Ecoinvent dataset [29]. Mass balance for chemical recycling (Scenarios 2–4) were derived from lab−scale experimental data (Table S3) reported in the literatures [21,30,31]. The energy requirements of the recycling strategies were evaluated across the entire downstream processing, encompassing both production reactions and purification. Total energy consumption was estimated using Aspen Plus™ process simulations, employing a series of distillation columns to achieve ternary separation of product, solvent, and unreacted nucleophile. Operating conditions for key equipment are detailed in Table S4. Background data for the LCA were sourced from the Ecoinvent v3.10 (cut−off system model) database [29].

2.3. Reference framework

The PC − PLA waste analysed in this study was considered to have been derived from LWP, and disposable cutlery, commonly used at public events such as festivals and recreational gatherings, which is generally directed to MSWI facilities. A conservative estimate assumed that PLA contributed the 98 % of the waste stream, with the remaining 2 % comprising typical municipal solid waste materials. For Scenario 5, which involves direct incineration, data were sourced from the Ecoinvent database, with the assumption that the waste consists solely of PLA with a net calorific value of 18 MJ/kg. PLA was considered to contain 0.84 kg of biogenic carbon per kg of material, as reported in prior research [32]. Incineration was modelled to occur in a waste−to−energy facility equipped with dry flue gas treatment and selective catalytic reduction to control NO_x emissions. It was assumed that 49 % of the PLA's heating value could be recovered, with energy distributed as 29 % electricity and 71 % steam. The recovered energy was credited with offsetting the average Australian electricity and district heating mix.

2.4. Handling multi−functionality

In LCA, multi−functionality is considered when there is a multiple output in a system or utilizes inputs derived from other product life cycles, requiring appropriate distribution of environmental impacts among co − products and shared processes. To ensure accurate and consistent modelling, allocation procedures must be applied at two levels: the process level, which includes the collection, sorting, and transportation, and the system level, where credits are assigned for useful outputs from end−of−life treatments such as recycling and energy recovery. In this study, each treatment scenarios used PC − PLA waste as the preliminary feedstock. The waste stream analysed consisted of PLA − based LWP and cutlery, collected and processed together with other mixed waste fractions. The analysis considered all relevant steps, from collection and sorting to final treatment, ensuring proper allocation of environmental burdens to PLA.

At the system level, the zero burden approach [33] was applied, meaning the waste entering the EoL processes carry no upstream burdens. Thus, only impacts and benefits from waste treatment were considered. In this context, credits were given for useful outputs that reduce the environmental burden elsewhere in the system. For instance, credits regarding the thermal treatment of PLA waste were assigned for electricity and heat generation, as these outputs replace conventional energy sources in the grid, thereby reducing fossil fuel consumption. Similarly, in the chemical recycling process, credits were awarded for producing rPLA, which substitutes virgin PLA production and significantly reduces environmental impacts associated with material extraction and synthesis.

2.5. Impact assessment methodology

The assessment methods used in this study encompass a comprehensive set of impact categories relevant to bioplastics. Global Warming Potential (GWP) was calculated over a 100 − year time period using characterization factors from the Intergovernmental Panel on Climate Change (IPCC) fifth assessment report [34]. The assessments were derived solely from impact factors provided by the Ecoinvent database version 3.10, without additional customized equation or calculations. The selected 16 impact categories included climate change (GWP), Acidification Potential (AP), Freshwater, Marine, and Terrestrial ecotoxicity (FET, MET, and TET), Fossil Energy Resources (FER), Freshwater and Marine Eutrophication (FWEP and MEP), Ionizing Radiation

(IR), Carcinogenic and Non − carcinogenic Human Toxicity (CHT and NCHT), Particulate Matter Formation (PMF), Photochemical Oxygen Formation (POF), Ozone Depletion (OD), Land Use (LU), and Water Use (WU).

2.6. Life cycle inventories

Life Cycle Inventory (LCI) is the phase of LCA study that quantifies all inputs (e.g., energy, materials) and outputs (e.g., emissions, waste) associated with a product system [35]. This section provides a comprehensive overview of the evaluated end−of−life scenarios for handling post−consumer PLA waste, including the associated life cycle inventory data. Key LCI data are summarized in Table 1, with comprehensive details available in the supplementary materials (Tables S6–S14).

2.6.1. Pre − treatment process

The municipal waste stream, comprising LWP and cutlery waste, contains various materials including iron, aluminium, different plastics, composites, cardboard, papers, and misthrows, with PLA as a key fraction (Table S5). The collected PLA waste was delivered to a treatment facility, where it underwent preliminary sorting using NIR technology [43], consuming 48 kWh per tonne of waste. This process yielded a relatively pure PLA waste stream, consisting of approximately 98.8 % PLA and 1.2 % impurities, which included small amounts of metals, plastics, paper, and other inactive materials.

To effectively remove residual impurities such as imprints and printing inks, which could not be eliminated during the initial sorting, the sorted fraction was washed with a 2 % aqueous NaOH solution at 70 °C and dried. Additional separation processes, such as shredding, airflow separation, and density − based washing, were carried out for the mechanical recycling. The dried PLA − rich fraction then underwent further purification through a zigzag classifier, electrostatic plastic−metal and plastic–plastic separation, effectively reducing impurities from an initial 25–30 % to around 5 %. In comparison, chemical recycling methods (Scenarios 2–4) were more contamination−tolerant, eliminating the need for these post−washing refinement steps. The higher tolerance for impurities in chemical recycling made it a more flexible approach for handling PC − PLA waste with variable contamination levels. For direct incineration (Scenario 5), no pre − treatment was required. It was assumed that PLA waste was transported directly from the collection site to the incineration facility. Collection and transport to the treatment facility were carried out using a diesel−powered truck. Detailed inventory data for the pre − treatment and transportation stages across all scenarios are presented in Table S6.

2.6.2. Mechanical recycling

In the LCI for mechanical recycling (Scenario 1), the primary energy input was electricity for the extrusion process, with an estimated consumption of approximately 207 kWh per tonne of PLA flakes. The electricity demand for subsequent steps such as melt sieving and pelletizing was considered negligible. Additional resource inputs included cooling water for temperature regulation during extrusion and pelletizing. The LCI data for mechanical recycling of PC − PLA is shown in Table S7.

2.6.3. Hydrolysis

The hydrolysis process at elevated temperatures is considered in scenario 2, utilizing a method developed by the Piemonte et al., (2013) [21]. This process yielded lactic acid (LA) as the primary product, which is subsequently polymerized to produce rPLA. The energy demand for the downstream processing of PLA waste recycling was quantified using process simulations in Aspen Plus™, focusing on the hydrolysis of PLA into lactic acid and its subsequent polymerization into rPLA. A detailed stream table, including key operational conditions, stream names, and mass flow data exported from Aspen Plus, is provided (Table S8) alongside the comprehensive modelling framework, as illustrated in Fig. S1.

The reaction was autocatalytic, achieving a conversion efficiency of approximately 1111 kg of lactic acid/t PLA waste which was then repolymerized to obtain 950 kg of rPLA. Energy and material inputs for hydrolysis, including steam, electricity, and catalysts, were accounted for, along with emissions and waste streams. The LCI for PLA recycling, detailing key process inputs and outputs, is provided in Table S9.

2.6.4. Methanolysis

The production of methyl lactate at 57 °C by methanolysis was involved in scenario 3, following the process developed by Anneaux et al., (2018) [30]. The energy requirements for downstream processing in PLA waste recycling were assessed through process simulations in Aspen Plus™, emphasizing the methanolysis of PLA into methyl lactate and its subsequent polymerization into rPLA. The comprehensive modelling framework is shown in Fig. S2 along with a detailed stream table (Table S10). This process achieved a yield of approximately 1311 kg of methyl lactate per FU, which was subsequently converted to lactic acid and further polymerized to rPLA.

The reaction utilized methanol, chloroform, and stannous octoate as the nucleophile, solvent and catalyst respectively. Energy and material inputs, including solvent recovery and catalyst efficiency, were considered in the LCI data. Emissions, waste streams, and byproducts from the process were also accounted for to ensure a comprehensive assessment. The detailed LCI data for PLA recycling through alcoholysis is provided in Table S11.

2.6.5. Ethanolysis

The production of ethyl lactate at 50 °C by ethanolysis was considered in scenario 4, based on the process developed by Majgaonkar et al., (2021) [31]. The energy requirements for downstream processing in PLA waste recycling were assessed through process simulations in Aspen Plus™, emphasizing the ethanolysis of PLA into ethyl lactate and its subsequent polymerization into rPLA. The detailed modelling framework is illustrated in Fig. S3 which included optimum operational conditions and mass flow rate throughout the entire downstream process (Table S12).

This process yielded approximately 1446 kg of ethyl lactate per FU. The reaction utilized ethanol, acetone, and zinc acetate as the nucleophile, solvent and catalyst respectively. The life cycle inventory accounted for key process inputs, including energy consumption, material use, emissions, and byproduct formation. The detailed inventory data for PLA recycling via ethyl lactate production is presented in Table S13.

2.6.6. Thermal treatment or incineration

A diesel−powered truck was used to transport the collected municipal waste from the collection point to the incineration facility. In this process (Scenario 5), a dataset representing a traditional incineration plant with dry flue gas treatment was used. The assumptions related to transport distance, collection time, and stop duration was aligned with those used during the preparation of PC − PLA waste. The recovered heat and electricity were assumed to offset energy from the average Australian heating grid, and the resulting potential benefits gained from displacing fossil fuel use were incorporated into the LCA as credit allocations. Datasets for assigning heat and electricity credits are provided in Table S14.

2.7. Sensitivity analysis

Sensitivity analysis was conducted to evaluate the influence of variations in model input values on the results. Key parameters included chemical consumption, electricity use, heat generation from natural gas, and diesel use in machinery and transport. Furthermore, the environmental implications of shifting entirely toward renewable energy were

Table 1
Key life cycle inventory data for the end−of−life treatment of 1 t PC − PLA waste.

Scenarios	Process	Unit	Quantity	Provider	Reference and remarks
Pre − treatment					
	Input flows				
	Fuel energy consumption	MJ	9.84	–	[36]
	Energy for machineries	kWh	6.3	Market for diesel, Cutoff, U	[37]
	Total energy demand for processing	kWh	206.83	Market for electricity, medium voltage, Cutoff, U	[38–42]
	Transport	km	200	Market for transport, freight, lorry >32 metric ton, EURO5, Cutoff, U	Estimated
	Output flows				
	Residues after sorting	kg	5	–	Measured
	Residues after washing	kg	9	–	Measured
	Residues after shredding	kg	6	–	Measured
	Net amount of PLA waste after treatment	kg	980	–	Calculated
Mechanical recycling (Scenario 1) [16]					
	Input flows				
	Electricity demand	kWh	207	Market for electricity, medium voltage, Cutoff, U	Estimated
	Water	L	20	Elementary flows/Resource/in water	Estimated
	Output flows				
	PLA loss	%	1	–	Estimated
	Mechanically rPLA	kg	970	–	Calculated
	Impurities	kg	50	–	Measured
	Wastewater	L	10	Market for wastewater, average, Cutoff, U − RoW	Estimated
	CO_2 emission	kg	7	Elementary flows/Emission to air/high population density	Estimated
Hydrolysis (Scenario 2) [21]					
	Input flows				
	Electricity demand for recycling	kWh	10	Market for electricity, medium voltage, Cutoff, U	Measured and scaled−up
	Solvent (water) demand for recycling	L	5000	Elementary flows/Resource/in water	Measured and scaled−up
	Thermal energy demand for the entire processing	MJ	24,208.32	Market for heat, district or industrial, natural gas, Cutoff, U	Aspen Plus™
	Output flows				
	Amount of lactic acid produced	kg	1110.78		Measured and scaled−up
	Amount of rPLA produced	kg	950	–	Measured and scaled−up
	Wastewater	L	3000	Market for wastewater, average, Cutoff, U − RoW	[29]
	CO_2 emission from processing	kg	20	Elementary flows/Emission to air/high population density	[29]
Methanolysis (Scenario 3) [30]					
	Input flows				
	Electricity demand for recycling	kWh	10	Market for electricity, medium voltage, Cutoff, U	Measured and scaled−up
	Solvent (chloroform) demand for recycling	kg	5612	Market for chloroform, Cutoff, U	Measured and scaled−up
	Nucleophile (methanol) demand for recycling	kg	2316	Market for methanol, Cutoff, U	Measured and scaled−up
	Catalyst [Tin(II) octate] demand for recycling	kg	21.56	–	Measured and scaled−up
	Thermal energy demand for the entire processing	MJ	20,072.6	market for heat, district or industrial, natural gas, Cutoff, U	Aspen Plus™
	Output flows				
	Amount of methyl lactate produced	kg	1311.36	–	Measured and scaled−up
	Amount of rPLA produced	kg	960	–	Measured and scaled−up
	CO_2 emission from processing	kg	22	Elementary flows/Emission to air/high population density	[29]
Ethanolysis (Scenario 4) [31]					
	Input flows				
	Electricity demand for recycling	kWh	10	market for electricity, medium voltage, Cutoff, U	Measured and scaled−up
	Solvent (acetone) demand for recycling	kg	5000	market for acetone, liquid, Cutoff, U	Measured and scaled−up
	Nucleophile (ethanol) demand for recycling	kg	2940	market for ethanol, without water, in 95 % solution state, from fermentation, Cutoff, U	Measured and scaled−up
	Catalyst [zinc acetate] demand for recycling	kg	19.6	–	Measured and scaled−up
	Thermal energy demand for the entire processing	MJ	12,208.35	market for heat, district or industrial, natural gas, Cutoff, U	Aspen Plus™
	Output flows				
	Amount of ethyl lactate produced	kg	1445.86	–	Measured and scaled−up
	Amount of rPLA produced	kg	946	–	Measured and scaled−up
	CO_2 emission from processing	kg	20	Elementary flows/Emission to air/high population density	[29]
Incineration (Scenario 5) [29]					
	Input flows				

(*continued on next page*)

Table 1 (*continued*)

Scenarios	Process	Unit	Quantity	Provider	Reference and remarks
	Municipal waste to incineration	kg	12	–	Calculated
	Chemical, inorganic	kg	3.18	market for chemical, inorganic, Cutoff, U	[29]
	Heat for energy	MJ	100.24	market for heat, district or industrial, natural gas, Cutoff, U	[29]
	Water, decarbonized	kg	538.77	market for water, decarbonised, Cutoff, U	[29]
	Oxygen	kg	1332.69	Elementary flows/Resource/in air	[29]
	Output flows				
	Credit for electricity produced from incineration	kWh	725.54	–	Calculated according to ecoinvent dataset
	Credit for energy produced from incineration	MJ	6390.33	–	Calculated according to ecoinvent dataset
	Direct CO_2 emission	kg	241.37	Elementary flows/Emission to air/high population density	[29]
	Biogenic CO_2 release	kg	3080	Elementary flows/Emission to air/unspecified	[32]
	CO emission	kg	23.07	Elementary flows/Emission to air/unspecified	[29]
	Wastewater	L	450	market for wastewater, Cutoff, U – RoW	[29]

analysed.

The selected parameters for sensitivity analysis were identified based on their contribution to total GWP. This analysis involves varying each parameter individually by ±30 % within the defined boundaries, using the one−at−a − time (OAT) method following standard practice outlined in ISO 14044:2006 [26]. For each variation a, calculation of a fresh impact value was established and benchmarked against the baseline within the corresponding impact category b [44]. The sensitivity coefficient (SC) was then determined as the ratio of absolute changes (Eq. 1) and sensitivity ratio (SR) as the ratio between the two relative changes (Eq. 2).

$$SC_a^b = \frac{(\Delta \text{ result})^b}{(\Delta \text{ parameter})_a} \approx \frac{\partial z_b}{\partial x_a} \tag{1}$$

$$SR_a^b = \frac{\left(\frac{\Delta \text{ result}}{\text{initial result}}\right)^b}{\left(\frac{\Delta \text{ parameter}}{\text{initial parameter}}\right)_a} \approx \frac{\partial z_b}{\partial x_a} \frac{x_a}{z_b} \tag{2}$$

In this equation, $a = 1, 2, 3\ldots$, n denotes the input parameters tested, and $b = 1, 2, 3\ldots$, m represents the impact metrics within the adopted characterization framework. The second expression follows the formulation by Heijungs and Lenzen [45], and in this formulation $z = z(x, y)$ denotes the output, x_a is the input parameter, and x_b refers to the impact metrics. For two input variables, x and y, the output variable z is expressed as a general function of both.

While the ±30 % parameter variation used in the sensitivity analysis reflects standard practice in LCA [26], it is limited by a lack of site−-specific variability data and assumes independent variation of inputs. This approach may overlook interactions between parameters, such as energy demand and solvent recovery efficiency. Additionally, the fixed variation range may not accurately represent real−world variability across different geographic or operational contexts.

2.8. Uncertainty analysis

Uncertainty analysis is a critical component of this study as it addresses the inherent result variability in data sources and enhances the robustness of our findings. This study used a stochastic approach for uncertainty propagation. The scenarios considered here, we assumed that all data−related uncertainties can be represented by lognormal distributions, which is commonly used to represent non − negative and positively skewed data points found in LCA [46]. In lognormal distributions, the square of the geometric standard deviation (σ_g^2) covers 95 % of the confidence interval and is often used as a quantitative indicator of uncertainty for a given parameter [44].

In this study, σ_g^2 was calculated as a function of basic and additional uncertainty factors (Eq. 3 and 4). Basic uncertainty factors (U_b) represent uncertainties associated with measurement errors and normal data fluctuations. For example, the basic uncertainty factor for CO_2 emission data ($U_b = 1.05$) is lower than that of heavy metal emission to water ($U_b = 5.00$) because CO_2 measurements are generally more precise and can be easily validated against the fuel inputs [47]. In our investigation, U_b value for each parameter was based on expert estimations provided by PRé Sustainability [47].

In this LCA modelling the applied clasical uncertainty factors ($U_{a,c}$) were determined using the Pedigree matrix (Eq. 4). This approach accounts for five uncertainty criteria, including data reliability, completeness, temporal correlation, geographical correlation, and other technological correlation [48]. Each criterion was given a score based on our assessment, which was then used to calculate its corresponding uncertainty factor [48]. The final uncertainty factor for each parameter (σ_g^2) was used to generate an uncertainty distribution. All calculated results related to the uncertainty analysis is provided in Table S15 −

Table 2
Environmental impact results for end−of−life treatment of 1 t of PC − PLA waste, calculated using the ReCiPe 2016 v1.03 Midpoint (H) method.

Impact categories	Units	Mechanical recycling (Scenario 1)			Hydrolysis (Scenario 2)			Methanolysis (Scenario 3)			Ethanolysis (Scenario 4)			Incineration (Scenario 5)		
		Emissions	Savings	Net	Emissions	Savings	Net	Emissions	Savings	Net	Emissions	Savings	Net	Emissions	Savings	Net
Climate change	kg CO_2 eq/FU	398	−3087	−2690	1293	−2193	−900	1616	−1870	−254	1411	−2075	−665	3404	−1771	1633
Acidification potential	kg SO_2 eq	1.17	−32.94	−31.77	1.68	−32.43	−30.75	2.32	−31.79	−29.47	6.06	−28.05	−21.99	15.34	−18.21	−2.31
Freshwater ecotoxicity	kg 1,4 − DCB eq	18.56	−461	−443	52.12	−428	−376	58.23	−422	−364	95.09	−385	−290	170	−301	−122
Marine ecotoxicity	kg 1,4 − DCB eq	26.33	−615	−588	74.71	−566	−492	85.68	−555	−470	113	−528	−416	250	−378	−116
Terrestrial ecotoxicity	kg 1,4 − DCB eq	1390	−45,041	−43,651	7627	−38,805	−31,179	10,305	−36,127	−25,822	17,091	−29,341	−12,250	19,444	−25,886	−5339
Fossil energy resources	kg oil eq	100	−780	−680	429	−452	−23.38	777	−103	674	349	−532	−183	671	−159	562
Freshwater eutrophication	kg PO_4 eq	0.57	−2.27	−1.7	0.35	−2.49	−2.14	0.41	−2.43	−2.02	0.52	−2.32	−1.80	1.17	−1.42	−0.002
Marine eutrophication	kg N eq	0.0371	−2.52	−2.49	0.085	−2.48	−2.39	0.09504	−2.47	−2.37	0.77	−1.79	−1.03	1.20	−1.34	−0.12
Carcinogenic human toxicity	1,4 − DCB eq	52.06	−798	−746	170	−680	−509	222	−628	−406	227	−623	−397	330	−493	−137
Non − carcinogenic human toxicity	1,4 − DCB eq	710	−9296	−8586	1092	−8915	−7823	1274	−8733	−7459	2175	−7832	−5657	3626	−6047	−2088
Ionizing radiation	kBq Co − 60 eq	1.25	−90.76	−89.51	9.85	−82.16	−72.31	15.23	−76.78	−61.55	16.54	−75.47	−58.93	39.14	−51.89	−11.77
Land use	m^2 × a crop eq	3.58	−1311	−1308	10.53	−1304	−1294	12.98	−1302	−1289	883	−432	452	53.67	−1259	−1202
Ozone depletion	kg CFC − 11 eq	2.91E−04	−3.25E−03	−2.96E−03	4.34E−04	−3.11E−03	−2.67E−03	1.17E−02	−8.20E−03	3.54E−03	3.60E−03	−6.44E−05	3.67E−03	7.88E−04	−2.62E−03	−1.69E−03
Particulate matter formation	kg PM2.5 eq	0.42	−10.66	−10.24	0.72	−10.36	−9.64	1.02	−10.06	−9.04	1.81	−9.27	−7.46	1.91	−8.96	−6.84
Photochemical oxygen formation	kg NO_x eq	0.8	−8.62	−7.82	1.79	−7.63	−5.84	2.46	−6.96	−4.50	3.04	−6.38	−3.34	2.70	−6.30	−3.18
Water use	m^3	0.8	−224	−223	2.40	−223	−220	3.42	−222	−218	95.90	−129	−33.2	168	−56.45	112

S19. Hence, data uncertainties were methodically propagated into result uncertainties using Monte Carlo analysis, during which multiple analysis iterations were performed to simulate a range of possible outcomes, thereby providing a probabilistic distribution of the LCA results. In this study, Monte Carlo analysis was repeated for 10,000 runs for each scenario. The resulting impact scores were analysed by creating error bar charts and probability histograms to reflect the uncertainty of the LCA results. To improve the legibility of these figures, extreme outliers were excluded. No more than 50 outliers were removed for each result.

$$\sigma_g^2 = \exp\sqrt{[\ln(U_b)]^2 + [\ln(U_{a,c})]^2} \quad (3)$$

$$U_{a,c} = \sum_{1}^{5} \sigma_n^2 \quad (4)$$

Where σ_g^2 represents the final uncertainty; U_b is the basic uncertainty factor; and $U_{a,c}$ represents classical additional uncertainty factor, which was determined using the pedigree matrix; σ_1^2, σ_2^2, σ_3^2, σ_4^2, and σ_5^2 represent pedigree matrix scores in data reliability, data completeness, temporal correlation, geographical correlation, and further technological correlation, respectively.

3. Result and discussion

According to ISO 14044:2006, relevant impact categories should address significant environmental issues aligned with the study's goal and scope, be scientifically robust, quantitative, and represent the environmental issues of the product system [49]. This research work considered 16 impact categories to quantify the environmental burdens associated with treating 1 t of PC − PLA waste across different EoL methods (Table 2).

3.1. Climate change

This LCA study evaluated the climate change impact of five EoL scenarios: mechanical recycling, hydrolysis, methanolysis, ethanolysis, and incineration, for 1 t PC − PLA waste, considering emissions from pretreatment stages as well as environmental credits from substituting virgin PLA with recycled products. The results revealed that all recycling−based options exhibited net negative GWP, indicating a climate benefit, while incineration remained a net emitter despite energy recovery.

3.1.1. Mechanical recycling

Each recycling process contributed to GWP through different mechanisms, primarily driven by energy consumption, chemical reactions, and waste generation. Mechanical recycling (Scenario 1), the least energy−intensive method, involved shredding, melting, and reprocessing. PC − PLA, generating minimal emissions (398.06 kg CO_2 eq/FU) primarily from electricity use (163.83 kWh/FU), which was responsible for 88.5 % of the CO_2 emission, and material handling (Fig. 2). Electricity grid−mix of Australia was considered in this study, which is largely generated from natural gas and coal, each contributing 17 % and 46 % respectively. Due to the dominance of these fossil fuels in the energy supply, their combustion results in substantial CO_2 emissions. The biogenic carbon was preserved within the polymer structure, preventing its release into the atmosphere and maintaining a low net climate impact. However, mechanical recycling of PC − PLA has several limitations in terms of rPLA quality and applicability despite its low energy consumption and environmental benefits. The process involves melting and reprocessing, which degrades the polymer chains, leading to reduced molecular weight and inferior mechanical properties in rPLA. Additionally, contamination from additives, colorants, and other polymers limits its purity and recyclability, making it unsuitable for high−performance applications. Repeated mechanical recycling further exacerbates these issues, necessitating blending with virgin PLA or alternative treatment methods to maintain product integrity [50].

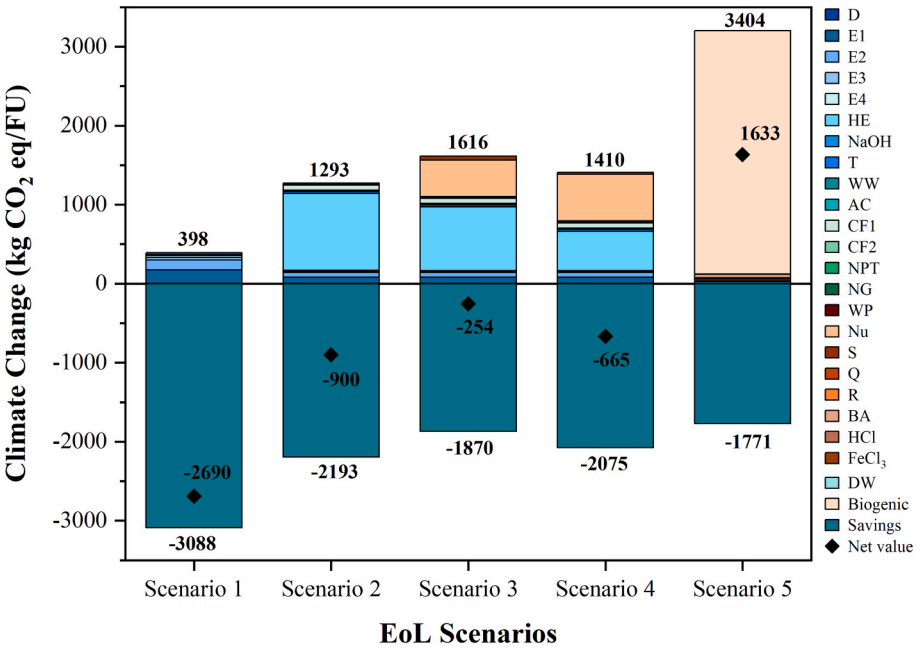


Fig. 2. Climate change impact for different scenarios of EoL options for PLA. Functional unit (FU) = 1 t PC − PLA waste; (D = market for diesel, E1 = electricity production from lignite, E2 = electricity production from hard coal, E3 = electricity production from natural gas, E4 = electricity production from other sources, HE = market for heat energy from natural gas, T = market for transport, WW = wastewater, AC = market for activated carbon, CF1 = market for organic chemical factory, CF2 = market for inorganic chemical factory, NPT = market for naphtha, NG = market for natural gas, WP = waste plastics, Nu = market for nucleophiles (methanol for scenario 3 and ethanol for scenario 4), S = market for solvent (chloroform for scenario 3 and acetone for scenario 4), Q = market for quicklime, R = solid residues, BA = bottom ash, DW = decarbonised water.

However, the scalability and regional applicability of mechanical recycling must be considered alongside its environmental profile. Owing to its lower capital and operational requirements, this method is particularly well−suited to small−to−medium enterprises and regions with limited technical infrastructure, such as in developing countries, where simple processing equipment and localized recycling streams are more feasible. Nonetheless, quality degradation limits its use in high−end applications unless enhanced with additional processing. In industrial settings, especially in developed countries, investments in purification can improve its viability.

3.1.2. Chemical recycling (hydrolysis vs alcoholysis)

Chemical recycling offers a significant advantage over mechanical recycling by breaking down PC − PLA into its monomers, allowing the production of high−purity rPLA with properties comparable to virgin PLA, without degradation in molecular weight or mechanical performance. Hydrolysis (Scenario 2) has a lower GWP of 1292.96 kg CO_2 eq/FU compared to other chemical recycling processes. This reduction is primarily due to the consumption of water, which reduces the reliance on other chemicals as a solvent that could have greater environmental impacts. However, hydrolysis requires a significant amount of heat energy due to the high temperatures needed for the reaction. This energy demand accounts for 75 % of the total GWP, making it a major contributor to the overall environmental impact of this process (Fig. 2). On the other hand, methanolysis (Scenario 3) and ethanolysis (Scenario 4) contributed to high GWP emissions, with values of 1616 kg CO_2 eq/FU and 1410 kg CO_2 eq/FU respectively. The primary cause of these elevated GWP emissions is the significant consumption of nucleophiles, as well as the considerable energy requirements for the entire downstream processing. When fresh methanol or ethanol is introduced into the alcoholysis process, the production of these nucleophiles must be accounted for in the background, further increasing the GWP. Specifically, the production of methanol accounted for 28 % of the total GWP, while ethanol production contributed 42 %. In terms of energy consumption, Scenario 3 required 805 MJ of energy, while scenario 4 needed 493 MJ which translated to 50 % and 35 % of the total GWP respectively (Fig. 2). The high energy usage was primarily due to the need for elevated temperatures and the maintenance of specific reaction conditions throughout the process, which resulted in increased emissions from energy production. Overall, scenario 3 was found to have the highest CO_2 emissions due to the elevated energy requirements and the incorporation of methanol, a fossil−based nucleophile. However, the considered recycling method exhibited a lower GWP compared to the production of virgin PLA from agricultural feedstocks, such as cane−sugar (3486 kg CO_2 eq/t PLA) [32], microalgae (19,176 kg CO_2 eq/t PLA) [32], corn stover (5790 kg CO_2 eq/t PLA) [51] and bamboo residues (3700 kg CO_2 eq/t PLA) [52]. Due to high capital investment, technical complexity, and energy requirements, these processes are more feasible in developed regions with access to clean energy, skilled labour, and advanced infrastructure. Large−scale industries may benefit from closed−loop recovery and higher material purity, justifying the energy and cost input. Conversely, in developing countries or small enterprises, the infrastructural and economic constraints may hinder the adoption of such technologies, unless regional capacity building and clean energy access is prioritized. Therefore, while chemically advanced, the practical deployment of these recycling routes remains scale− and region−dependent, requiring a careful balance between environmental gains and socio−technical feasibility.

3.1.3. Biogenic carbon release during incineration

The thermal treatment or incineration, unlike recycling methods, resulted in a substantial carbon footprint (3403 kg CO_2 eq/FU) due to the combustion of PC − PLA (Fig. 2). The process generated CO_2 emissions from two primary sources which are (i) the oxidation of the polymer backbone and (ii) the energy required for high−temperature treatment. Most critically, incineration led to the complete release of the biogenic carbon stored in PLA, which was considered 3080 kg CO_2 eq/t PLA in our previous study [32], significantly contributing to atmospheric CO_2 levels.

In the expanded system, credits or avoided burdens were accounted for by considering the substitution of virgin PLA with the rPLA obtained from each investigated scenario. Among the evaluated options, Scenario 1 demonstrated the highest GWP savings, reaching −2982.95 kg CO_2 eq/FU, primarily due to its low energy demand and minimal emissions. Chemical recycling methods, despite requiring higher energy input, still achieved significant GWP reductions by preserving biogenic carbon within the material cycle, with Scenario 2, Scenario 3, and Scenario 4 showing net savings of −2088 kg CO_2 eq/FU, −1765 kg CO_2 eq/FU, and − 1970 kg CO_2 eq/FU, respectively. In contrast, Scenario 5, which involved incineration of PC − PLA waste, resulted in substantial carbon emissions of 3403 kg CO_2 eq/FU. Although energy recovery from incineration provided some offset, leading to a GWP savings of −1771 kg CO_2 eq/FU, the overall impact remained strongly positive, highlighting its inefficiency in mitigating climate change. This comparison underscores the environmental benefits of recycling−based approaches over incineration in achieving a more sustainable end−of−life management strategy for PC − PLA. Nevertheless, the findings of Aryan et al., (2021) and Maga et al., (2019) support this result, demonstrating that the net GWP of recycling is lower than that of incineration [16,23]. However, variations exist in the outcomes, as their studies accounted for the replacement of equivalent reference products, such as lactic acid, methyl lactate, and ethyl lactate, rather than incorporating the entire downstream processing required to produce rPLA.

3.2. Comparative evaluation of impact assessment results

The comparative evaluation of the environmental burden for all five EoL treatment scenarios are illustrated in Fig. S4 and further summarized in a heat map (Table 3), where the minimum and maximum environmental impacts for each impact category are highlighted in green and red, respectively.

Among these, scenario 1 (mechanical recycling) demonstrated the lowest environmental impacts across most categories, primarily due to its minimal energy and chemical requirements. The extrusion phase, involving melting and pelletizing, accounted for over 85 % of the gross GWP and fossil energy use, largely due to electricity consumption. Upstream processes such as washing, shredding, and air separation also contributed to specific impact categories like PMF and WU through auxiliary energy and freshwater inputs. Despite these burdens, the substitution of virgin PLA with recyclate led to net − negative impacts across all assessed categories (Fig. 3), and the process exhibited minimal contributions to human toxicity and ecotoxicity indicators. However, mechanical recycling is not a long−term solution, as repeated cycles degrade PLA polymer quality, limiting its use in high−performance applications. Moreover, the process demands a high−purity PLA feedstock, making contamination a critical barrier to its scalability in large−scale circular economy systems. Nonetheless, mechanical recycling remains a highly effective and sustainable strategy for small−to−medium−scale operations with access to clean PLA waste streams.

Scenarios 2–4 exhibited comparatively higher environmental impacts across all categories due to their energy−intensive operations and significant reagent consumption. Among them, hydrolysis (Scenario 2) had relatively lower impacts than alcoholysis (Scenarios 3 and 4), largely because it is an autocatalytic process that uses water as a reactant, eliminating the need for external alcohols and catalysts that typically increase emissions and toxicity. However, hydrolysis faces notable operational challenges, including slow reaction kinetics, prolonged processing times, and lower yields of high−purity monomers, which necessitate additional purification steps. These inefficiencies elevate overall energy consumption, reducing its sustainability advantage. The process also demands high thermal energy input, primarily from natural gas−based systems, for depolymerization and solvent removal,

Table 3
Heat map of LCA results across all investigated scenarios.

Impact categories	Unit	Scenario 1	Scenario 2	Scenario 3	Scenario 4	Scenario 5
GWP	kg CO_2 eq/FU	−2690	−900	−254	−665	1633
AP	kg SO_2 eq	−31.77	−30.75	−29.47	−21.99	−2.31
FET	kg 1,4−DCB eq	−443	−376	−364	−290	−122
MET	kg 1,4−DCB eq	−588	−492	−470	−416	−116
TET	kg 1,4−DCB eq	−43651	−31179	−25822	−12250	−5339
FER	kg oil eq	−680	−23.38	674	−183	562
FWEP	kg PO_4 eq	−1.70	−2.14	−2.02	−1.80	−0.002
MEP	kg N eq	−2.49	−2.39	−2.37	−1.03	−0.12
CHT	kg 1,4−DCB eq	−746	−509	−406	−397	−137
NCHT	kg 1,4−DCB eq	−8586	−7823	−7459	−5657	−2088
IR	kBq Co−60 eq	−89.51	−72.31	−61.55	−58.93	−11.77
LU	m^2 × a crop eq	−1308	−1294	−1289	452	−1202
OD	kg CFC−11−eq	−2.96E−03	−2.67E−03	3.54E−03	3.67E−03	−1.69E−03
PMF	kg PM2.5 eq	−10.24	−9.64	−9.04	−7.46	−6.84
POF	kg NO_x eq	−7.82	−5.84	−4.50	−3.34	−3.18
WU	m^3	−223	−220	−218	−33.20	112

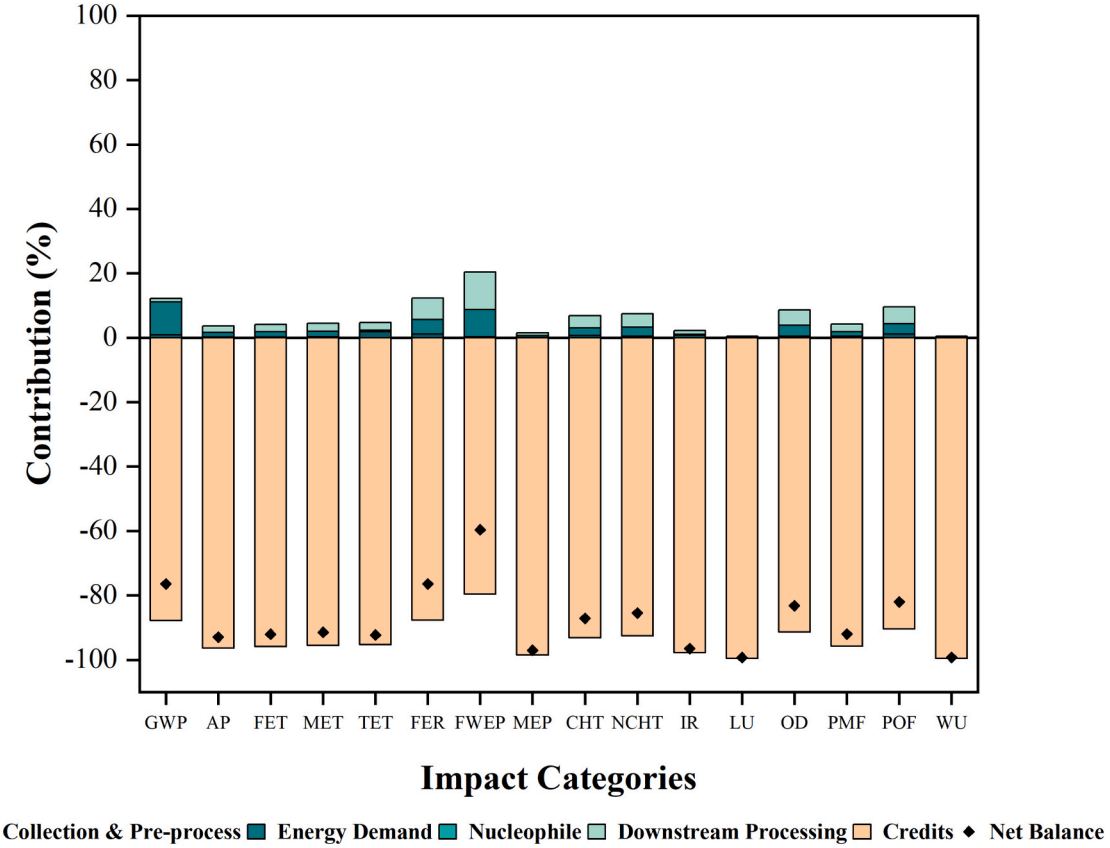


Fig. 3. Contribution analysis of mechanical recycling of PC − PLA waste (GWP = global warming potential; AP = acidification potential; FET = freshwater ecotoxicity; MET = marine ecotoxicity; TET = terrestrial ecotoxicity; FER = fossil energy resources; FWEP = freshwater eutrophication potential; MEP = .marine eutrophication potential; CHT = carcinogenic human toxicity; NCHT = non − carcinogenic human toxicity; IR = ionizing radiation; LU = land use; OD = ozone depletion; PMF = particulate matter formation; POF = photochemical oxygen formation; WU = water use).

contributing significantly to GWP (37.68 %) and FER (49.24 %) (Fig. 4). Despite these burdens, hydrolysis supports closed−loop recycling by recovering lactic acid for repolymerization into rPLA, thereby avoiding impacts from incineration or landfill disposal. This substitution leads to substantial emission offsets, achieving a net GWP of −900 kg CO_2 eq/FU and reinforcing hydrolysis's role in advancing circular economy principles, even amid performance limitations.

Alcoholysis, encompassing methanolysis (Scenario 3) and ethanolysis (Scenario 4), enables efficient monomer recovery with preserved material quality, though both routes present distinct environmental trade−offs. Methanolysis achieved substantial net reductions across most impact categories (Fig. 5); however, it exhibited elevated impacts in global warming potential (−253.48 kg CO_2 eq/FU), fossil energy resource use (674.02 kg oil eq/FU), and ozone depletion (3.54E−03 kg CFC − 11 eq/FU) (Table 3), primarily due to its dependence on fossil−−derived methanol and the use of chloroform as a halogenated solvent.

In contrast, ethanolysis demonstrated lower fossil fuel dependency (Fig. 6), reflected in a negative FER value of −182.46 kg oil eq/FU and a greater GWP reduction of −664.68 kg CO_2 eq/FU, attributed to the biogenic origin of ethanol. However, it incurred higher impacts in acidification, marine eutrophication, land use (451.56 m^2 × a crop eq/FU), water use (−33.20 m^3/FU), and ozone depletion (3.67E−03 kg CFC − 11 eq/FU), driven by the resource−intensive nature of agricultural feedstock cultivation and energy−demanding refining processes (Table 3). Ethanolysis also faced challenges in reaction efficiency and purification, as ethyl lactate is more complex to separate than methyl lactate, increasing downstream energy demand. Despite these burdens, both alcoholysis routes enabled the production of high−purity rPLA and delivered net environmental savings in most categories, underscoring their potential within circular economy frameworks when aligned with appropriate feedstock and energy sourcing strategies.

Incineration (Scenario 5) imposed the highest environmental burdens across multiple impact categories, primarily due to direct emissions from combustion and the destruction of material value. The combustion of PC − PLA waste released the full biogenic carbon content of PLA, resulting in substantial carbon emissions that exceeded the offsets from electricity and heat generation (Fig. 7). Elevated WU was also observed, mainly due to steam losses during boiler operation. Additionally, incineration eliminated the embedded fossil−based energy in PLA, requiring the production of new virgin PLA to meet demand, thereby intensifying FER impacts. The use of auxiliary fuels further increased fossil fuel dependency. While incineration provided some environmental credits through energy recovery, these were insufficient to counterbalance its overall negative environmental footprint, as clearly reflected in the heat map (Table 3), which highlights incineration in red across several categories. Compared to recycling−based strategies, incineration proved the least favourable end−of−life option for PLA waste, both environmentally and energetically.

3.3. Sensitivity analysis

Fluctuations in environmental burdens due to variations in critical input parameters were highlighted through the sensitivity analysis and the detailed results of this analysis are presented in Table 4. The proportional variation in output resulting from a percentage variation in the input parameters are indicated by the SR which helps to identify those with a high sensitivity. Conversely, the SC reflects the absolute output change per unit input change, showing direct impacts in real units. Even if an input is not highly significant, a high SC implies that small alterations in it can notably affect the outcome. However, SC values should be compared with caution due to their varying units, unlike SRs, which are unitless and enable consistent comparison across scenarios and categories.

It is indicated by the sensitivity analysis that energy−related factors

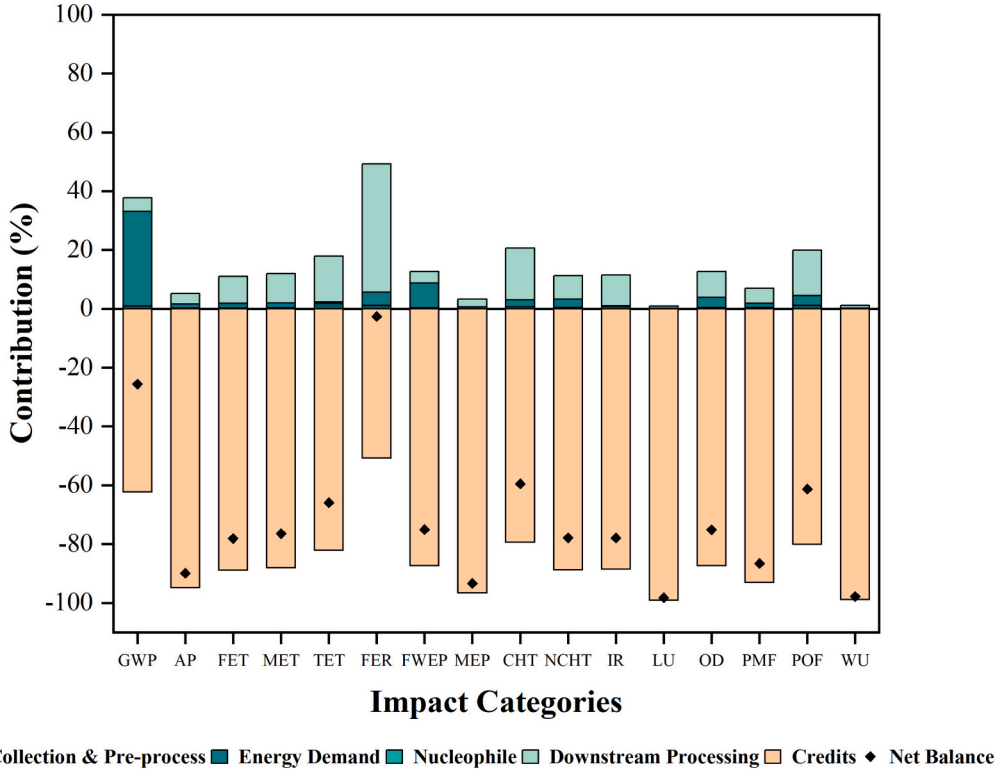


Fig. 4. Contribution analysis of hydrolysis of PC − PLA waste (GWP = global warming potential; AP = acidification potential; FET = freshwater ecotoxicity; MET = marine ecotoxicity; TET = terrestrial ecotoxicity; FER = fossil energy resources; FWEP = freshwater eutrophication potential; MEP = .marine eutrophication potential; CHT = carcinogenic human toxicity; NCHT = non − carcinogenic human toxicity; IR = ionizing radiation; LU = land use; OD = ozone depletion; PMF = particulate matter formation; POF = photochemical oxygen formation; WU = water use).

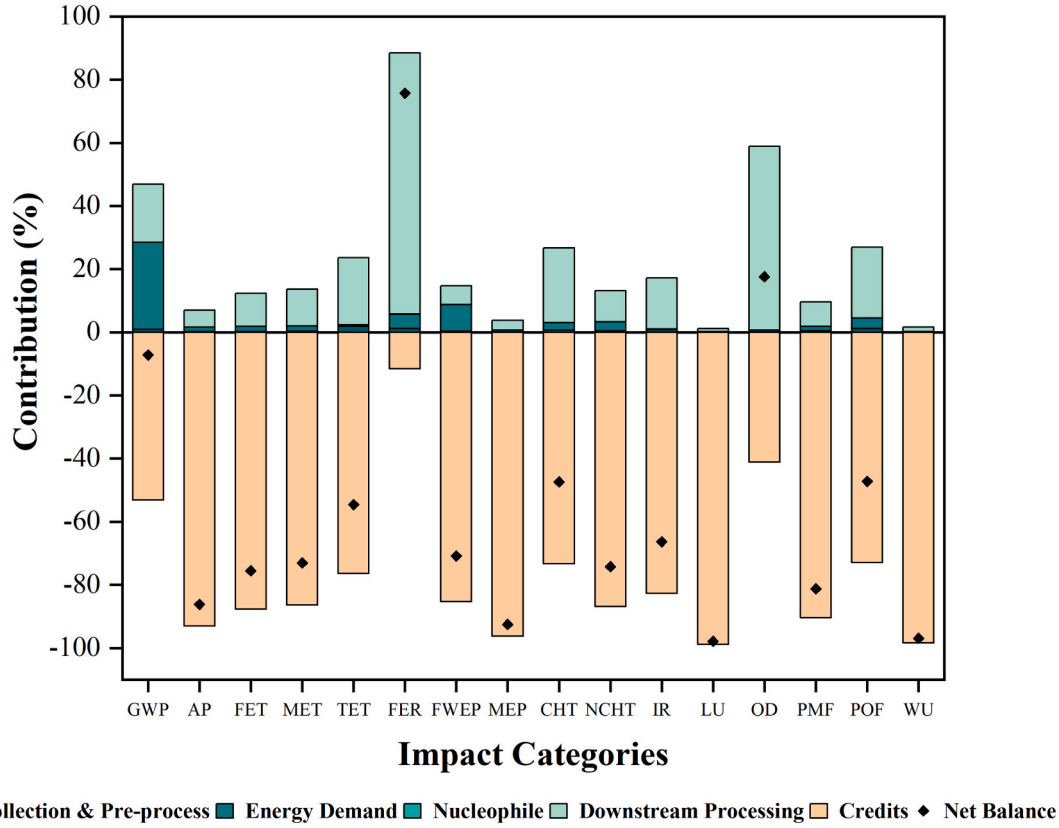


Fig. 5. Contribution analysis of methanolysis of PC − PLA waste (GWP = global warming potential; AP = acidification potential; FET = freshwater ecotoxicity; MET = marine ecotoxicity; TET = terrestrial ecotoxicity; FER = fossil energy resources; FWEP = freshwater eutrophication potential; MEP = .marine eutrophication potential; CHT = carcinogenic human toxicity; NCHT = non − carcinogenic human toxicity; IR = ionizing radiation; LU = land use; OD = ozone depletion; PMF = particulate matter formation; POF = photochemical oxygen formation; WU = water use).

strongly affect the CO_2 emissions—specifically electricity, heat, and diesel combustion—and by the use of chemicals such as solvents and nucleophiles across all EoL scenarios (Fig. 8). This is evidenced by the high SR and SC values associated with these parameters, indicating their substantial contribution to the overall environmental impact (Table 4).

GWP across all EoL pathways was primarily driven by energy consumption, with varying degrees of sensitivity depending on the process. Mechanical recycling exhibited a strong reliance on electricity for grinding, extrusion, and reprocessing, where a 30 % reduction in electricity consumption resulted in a 27 % decrease in GWP (from 398.06 to 292.38 kg CO_2 eq/FU), highlighting the potential of energy−efficient technologies in reducing its environmental impact. For hydrolysis, GWP was highly sensitive to heat consumption due to the elevated temperatures required for depolymerization. A 30 % reduction in heat usage led to a 23 % decrease in GWP, demonstrating the importance of thermal energy optimization in improving process sustainability. In alcoholysis, GWP was influenced not only by energy and electricity demand but also by the chemical−intensive nature of the process. Beyond depolymerization, significant energy is required for monomer purification, separation, and repolymerization, with key steps like reactor operation, distillation, crystallization, and polymerization driving emissions. A 30 % reduction in energy consumption across these downstream stages resulted in an estimated 15 % decrease in total GWP, emphasizing the need for process efficiency improvements. Additionally, solvent use played a crucial role in GWP sensitivity. By enhancing solvent recovery systems, GWP could be reduced by 13 %, primarily by decreasing reliance on fresh solvent production and minimizing disposal−related emissions. Efficient recovery not only reduces upstream emissions from solvent manufacturing and transportation but also lowers energy consumption in distillation and purification, making solvolysis a more sustainable PLA recycling approach. Incineration, however, exhibited the highest GWP among all EoL options, primarily due to direct CO_2 emissions from PLA combustion. While inherently carbon−intensive, some of the emissions associated with PLA incineration could be alleviated by improving waste−to−energy technology.

Optimizing the processes involved in EoL options for PLA recycling can significantly reduce not only GWP but also other environmental impacts such as AP, FET and CHT. By improving energy efficiency, for instance, less energy consumption results in fewer emissions of pollutants such as SO_2 and NO_x, which are responsible for acidification. This leads to a reduction in AP, particularly in processes like hydrolysis and incineration. Similarly, optimizing chemical usage, such as enhanced solvent recovery or reducing the use of harmful reagents, can lower the release of toxic substances into water bodies, decreasing freshwater ecotoxicity. For example, reducing the need for fresh solvent production not only reduces energy use but also mitigates harmful discharges into water systems. Furthermore, optimizing downstream operations like solvent recovery and purification can lead to reduced exposure to carcinogenic and other toxic chemicals, thus lowering the human toxicity impact.

The study found that global warming potential (GWP), along with acidification and eutrophication impacts, is largely driven by fossil fuel−based electricity generation, given the high energy intensity of PLA recycling processes. To quantify the influence of energy transition on GWP, a sensitivity analysis was performed by varying the proportion of renewable energy in the electricity mix at 20 %, 50 %, and 80 %. The results (Fig. 9) show a progressive decline in GWP as the share of renewable energy increases, highlighting the direct correlation between cleaner energy inputs and reduced carbon emissions.

In addition to these partial transition scenarios, a separate analysis

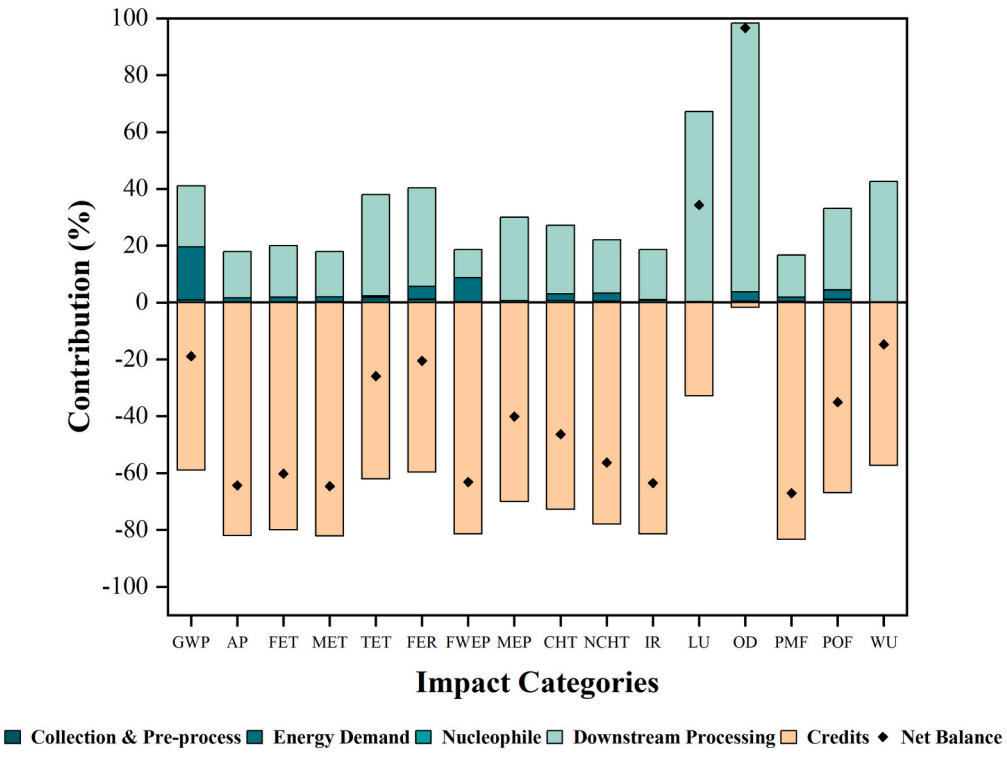


Fig. 6. Contribution analysis of ethanolysis of PC − PLA waste (GWP = global warming potential; AP = acidification potential; FET = freshwater ecotoxicity; MET = marine ecotoxicity; TET = terrestrial ecotoxicity; FER = fossil energy resources; FWEP = freshwater eutrophication potential; MEP = .marine eutrophication potential; CHT = carcinogenic human toxicity; NCHT = non − carcinogenic human toxicity; IR = ionizing radiation; LU = land use; OD = ozone depletion; PMF = particulate matter formation; POF = photochemical oxygen formation; WU = water use).

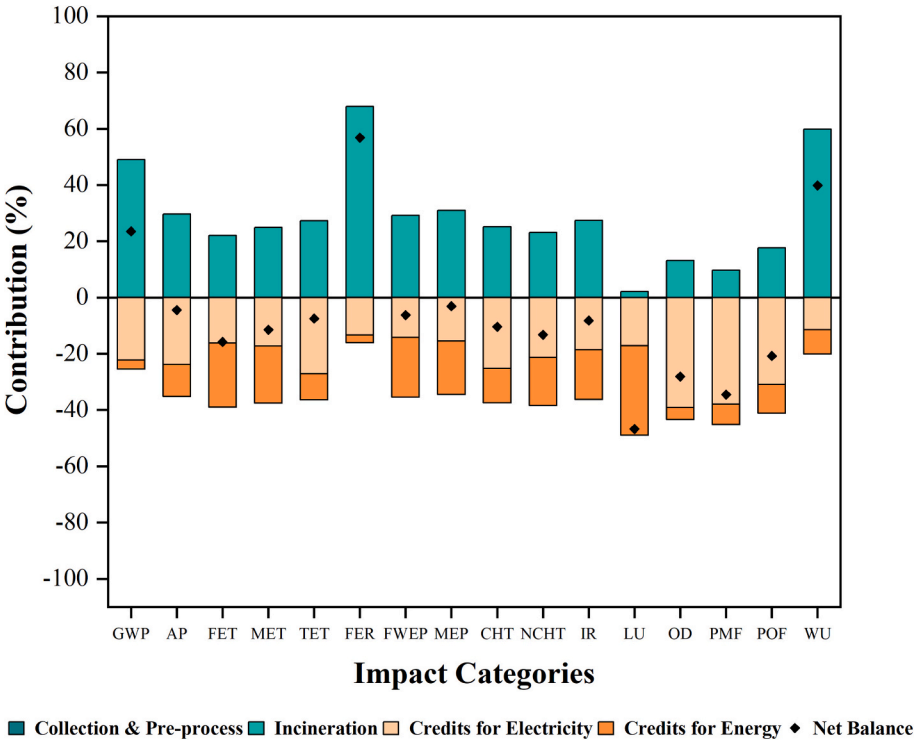


Fig. 7. Contribution analysis of ethanolysis of PC − PLA waste (GWP = global warming potential; AP = acidification potential; FET = freshwater ecotoxicity; MET = marine ecotoxicity; TET = terrestrial ecotoxicity; FER = fossil energy resources; FWEP = freshwater eutrophication potential; MEP = .marine eutrophication potential; CHT = carcinogenic human toxicity; NCHT = non − carcinogenic human toxicity; IR = ionizing radiation; LU = land use; OD = ozone depletion; PMF = particulate matter formation; POF = photochemical oxygen formation; WU = water use).

Table 4
Sensitivity analysis to select highly sensitive parameters across the EoL options for PC − PLA waste and their corresponding key impact categories.

Parameters	GWP			AP			FET			CHT		
	Sensitivity ratio(−)	Sensitivity coefficient (kg CO_2 eq/unit)	Impact range (±30 %)	Sensitivity ratio (−)	Sensitivity coefficient (kg SO_2 eq/unit)	Impact range (±30 %)	Sensitivity ratio (−)	Sensitivity coefficient (kg 1,4 − DCB eq/unit)	Impact range (±30 %)	Sensitivity ratio (−)	Sensitivity coefficient (kg 1,4 − DCB eq/unit)	Impact range (±30 %)
Mechanical Recycling												
Diesel burning	1.13E−02	1.38E−01	397–399	2.82E−02	1.03E−03	1.17–1.18	1.26E−02	7.18E−03	18.51–18.63	2.31E−02	3.69E−02	51.69–52.42
Electricity	8.85E−01	9.39E−01	292–504	9.04E−01	2.84E−03	0.85–1.50	9.43E−01	4.67E−02	13.33–23.81	8.73E−01	1.21E−01	38.41–65.69
Heat	1.55E−02	3.98E−02	396–400	2.82E−02	2.15E−04	1.18–1.19	3.59E−03	4.31E−04	18.56–18.58	6.40E−03	2.15E−03	51.95–52.16
NaOH	1.53E−02	1.53	396–398	2.82E−02	8.33E−03	1.17–1.18	1.80E−02	8.33E−02	18.47–18.66	2.24E−02	2.92E−01	51.70–52.41
Transport	5.54E−02	1.10E−01	391–405	2.82E−02	1.67E−04	1.16–1.19	3.23E−02	3.00E−03	18.40–18.74	7.36E−02	1.92E−02	50.90–53.21
Chemical Recycling (Hydrolysis)												
Diesel burning	3.48E−03	1.38E−01	1292–1294	1.98E−02	1.03E−03	1.68–1.69	3.84E−03	6.15E−03	52.07–52.18	7.05E−03	3.69E−02	170–171
Electricity	1.26E−01	9.39E−01	1244–1342	3.17E−01	3.07E−03	1.53–1.84	1.55E−01	4.66E−02	49.7–54.55	1.24E−01	1.21E−01	164–177
Energy	7.51E−01	3.98E−02	1002–1584	2.98E−01	2.05E−05	1.54–1.83	8.63E−02	1.85E−04	50.78–53.47	3.14E−01	2.20E−03	154–186
NaOH	4.72E−03	1.53	1291–1295	1.98E−02	8.33E−03	1.68–1.69	6.40E−03	8.33E−02	52.03–52.22	6.85E−03	2.92E−01	170–171
Transport	1.70E−02	1.10E−01	1286–1300	3.97E−02	3.33E−04	1.67–1.70	1.09E−02	2.83E−03	51.96–52.29	2.25E−02	1.92E−02	169–171
Activated carbon	1.32E−02	3.42	1288–1298	5.95E−02	2.00E−02	1.66–1.71	6.40E−03	6.67E−02	52.03–52.22	8.42E−03	2.87E−01	170–171
Naphtha	2.94E−03	7.92E−01	1292–1294	1.98E−02	6.94E−03	1.68–1.69	1.28E−03	1.39E−02	52.11–52.14	3.33E−03	1.18E−01	170–171
Natural gas	2.11E−03	7.59E−01	1292–1294	1.98E−02	9.26E−03	1.68–1.69	6.40E−04	9.26E−03	52.12–52.13	2.94E−03	1.39E−01	170–171
Chemical Recycling (Solvent: Methanol)												
Diesel burning	2.78E−03	1.38E−01	1615–1618	1.44E−02	1.03E−03	2.32–2.33	3.43E−03	6.15E−03	58.18–58.29	5.41E−03	3.69E−02	222–223
Electricity	1.01E−01	9.39E−01	1567–1665	2.30E−01	3.07E−03	2.17–2.48	1.39E−01	4.66E−02	55.81–60.66	9.49E−02	1.21E−01	216–228
Energy	4.99E−01	3.98E−02	1375–1858	1.87E−01	2.14E−05	2.00–2.45	6.41E−02	1.85E−04	57.11–59.35	2.00E−01	2.20E−03	209–235
NaOH	3.79E−03	1.53	1614–1618	1.44E−02	8.33E−03	2.32–2.33	5.72E−03	8.33E−02	58.14–58.33	5.26E−03	2.92E−01	222–223
Transport	1.36E−02	1.10E−01	1610–1623	2.87E−02	3.33E−04	2.31–2.34	9.73E−03	2.83E−03	58.06–58.40	1.73E−02	1.92E−02	221–223
Methanol	2.84E−01	1.00	1479–1754	2.87E−01	1.45E−03	2.13–2.52	1.10E−01	1.40E−02	56.32–60.15	2.39E−01	1.16E−01	206–238
Chloroform	3.25E−02	1.57E+08	1597–1635	4.31E−02	7.94E−03	2.30–2.35	2.06E−02	9.52E−02	57.88–58.59	3.53E−02	6.22E−01	220–224
Activated carbon	1.06E−02	4.17	1601–1632	4.31E−02	2.00E−02	2.30–2.35	5.15E−03	6.00E−02	58.14–58.32	6.46E−03	2.87E−01	222–223
Naphtha	2.37E−03	7.99E−01	1615–1617	1.44E−02	6.94E−03	2.32–2.33	1.14E−03	1.39E−02	58.22–58.25	2.55E−03	1.18E−01	221–222
Natural gas	1.69E−03	7.59E−01	1615–1617	1.44E−02	9.26E−03	2.32–2.33	5.72E−04	9.26E−03	58.23–58.24	2.40E−03	1.48E−01	221–222
Chemical Recycling (Solvent: Ethanol)												
Diesel burning	3.19E−03	1.38E−01	1409–1412	5.50E−03	1.03E−03	6.05–6.06	2.10E−03	6.15E−03	95.03–95.15	5.29E−03	3.69E−02	226–227
Electricity	1.16E−01	9.39E−01	1362–1460	8.25E−02	2.88E−03	5.91–6.21	8.52E−02	4.66E−02	92.66–97.52	9.29E−02	1.21E−01	220–233
Energy	3.49E−01	3.98E−02	1263–1558	3.85E−02	1.89E−05	5.98–6.13	2.42E−02	1.85E−04	94.41–95.78	1.20E−01	2.20E−03	219–235
NaOH	4.32E−03	1.52	1409–1413	5.50E−03	8.33E−03	6.05–6.06	3.51E−03	8.33E−02	95.00–95.19	5.15E−03	2.92E−01	226–227
Transport	1.56E−02	1.10E−01	1404–1417	5.50E−03	1.67E−04	6.05–6.07	5.96E−03	2.83E−03	94.92–95.26	1.69E−02	1.92E−02	226–228
Ethanol	4.19E−01	8.87E−01	1233–1588	7.48E−01	6.81E−03	4.69–7.42	4.64E−01	6.62E−02	81.87–108.32	3.45E−01	1.17E−01	203–250
Acetone	1.74E−02	2.21	1403–1418	1.10E−02	6.01E−03	6.04–6.08	1.09E−02	9.31E−02	94.79–95.40	1.87E−02	3.81E−01	225–228
Activated carbon	1.21E−02	3.42	1406–1416	1.10E−02	1.33E−02	6.03–6.08	3.15E−03	6.00E−02	95.00–95.18	6.32E−03	2.87E−01	226–227
Naphtha	2.72E−03	7.99E−01	1410–1412	5.50E−03	6.94E−03	6.06–6.07	3.51E−04	6.94E−03	95.08–95.10	2.50E−03	1.18E−01	226–227
Natural gas	1.94E−03	7.59E−01	1410–1412	5.50E−03	9.26E−03	6.06–6.07	3.51E−04	9.26E−03	95.08–95.10	2.35E−03	1.48E−01	226–227
Incineration (Thermal treatment)												
NaOH	7.04E−03	1.52	3396–3411	6.54E−02	6.35E−03	15.10–15.60	7.44E−03	8.05E−02	170–171	1.42E−02	2.99E−01	329–332
Quicklime	6.57E−03	1.19	3397–3410	2.18E−02	1.77E−03	15.30–15.40	9.79E−04	8.87E−03	170–170	1.31E−03	2.30E−02	330–330
Inorganic chemical	1.82E−03	1.95	3402–3406	2.18E−02	1.05E−02	15.10–15.40	3.33E−03	1.78E−01	170–171	5.65E−03	5.87E−01	330–331
Heat	1.18E−03	3.99E−02	3402–3405	2.18E−02	6.98E−03	15.30–15.40	1.96E−04	3.33E−04	170–171	7.07E−04	2.33E−03	330–331
HCl	6.76E−04	1.00	3403–3404	2.18E−02	1.46E−02	15.30–15.40	1.37E−03	1.02E−01	170–171	1.51E−03	2.18E−01	330–331
$FeCl_3$	5.29E−04	8.91E−01	3403–3404	2.18E−02	1.65E−02	15.30–15.40	1.57E−03	1.32E−01	170–171	2.22E−03	3.63E−01	330–331

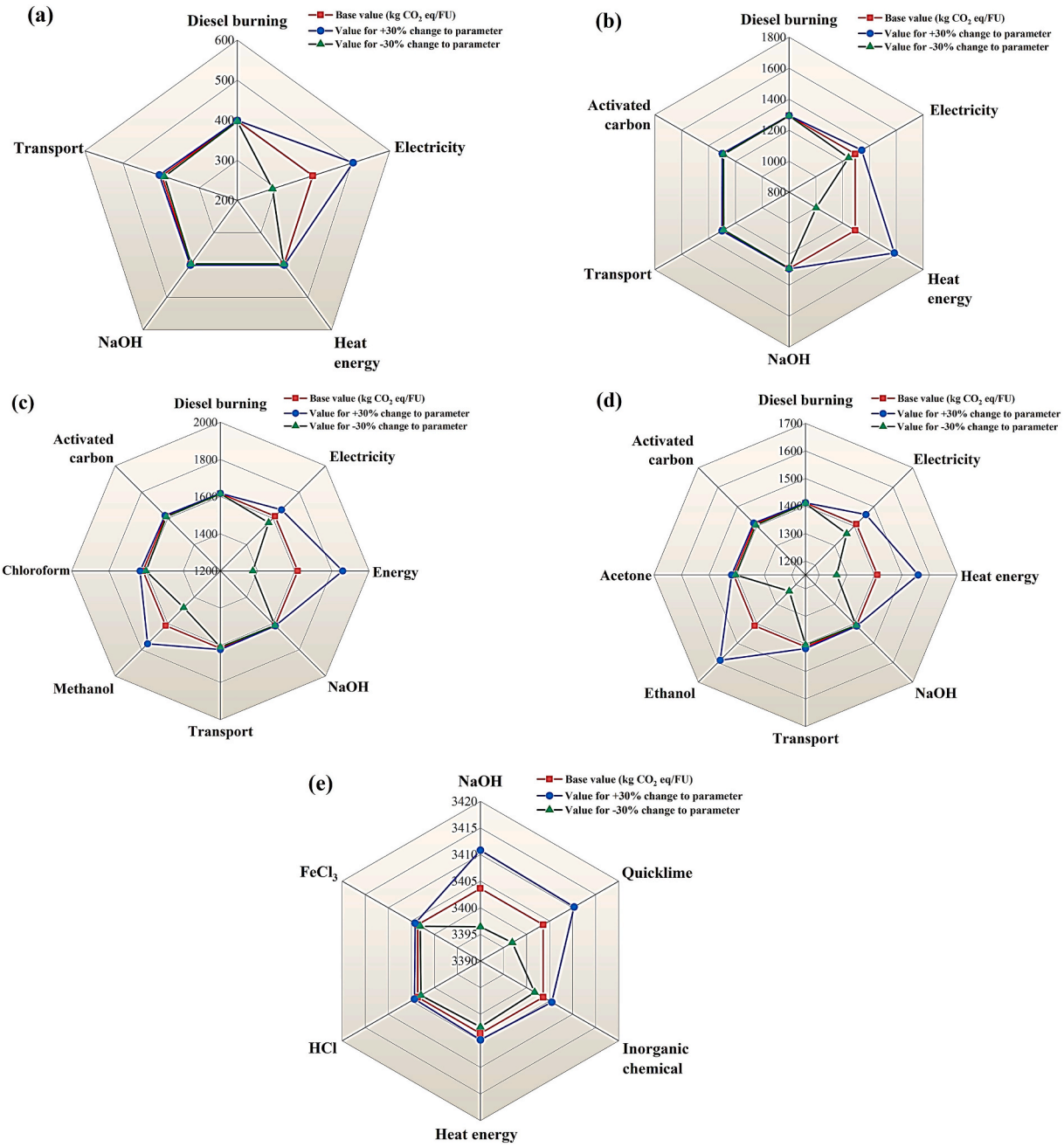


Fig. 8. Sensitivity analysis of climate change to the OAT of the parameters by ±30 % for the EoL options for PC − PLA waste (a) mechanical recycling, (b) hydrolysis, (c) alcoholysis (methanol), (d) alcoholysis (ethanol) and (e) incineration (FU = 1 t PC − PLA waste).

was conducted for a complete shift to 100 % renewable energy (wind, solar, and hydro), illustrated in Fig. 10. This scenario demonstrated an 84 % reduction in GWP compared to the fossil−based baseline, emphasizing the substantial environmental benefit of full decarbonization. Collectively, these findings underscore the critical role of renewable energy integration in minimizing the climate impacts of end−of−life PLA management and supporting a low−carbon circular economy. Beyond climate impact, this shift also holds the potential to mitigate other environmental burdens, such as acidification and eutrophication, by eliminating sulphur dioxide and nitrogen oxide emissions associated with fossil fuel combustion. The findings emphasize that integrating renewable energy into recycling processes could make PLA waste management significantly more sustainable. However, multiple factors affect the viability of this transition, with renewable energy infrastructure availability, economic viability, and regional energy policies being the key considerations. While the environmental benefits are clear, large−scale implementation would require strategic planning, investment, and policy support to ensure a smooth transition toward a low−carbon, resource−efficient recycling system.

3.4. Uncertainty analysis

Uncertainty analysis is a critical component of LCA, as it helps identify the limitations of the assessment and supports more informed decision−making. The results of the Monte Carlo simulations are presented in Fig. 11 as error bar charts, depicting the initially calculated impact scores alongside the propagated means, medians, and 95 % confidence intervals, which represent result variability. Additionally, the relative frequency histograms for the GWP category are illustrated in Fig. 12 and an additional LCIA and Monte Carlo results for all scenarios

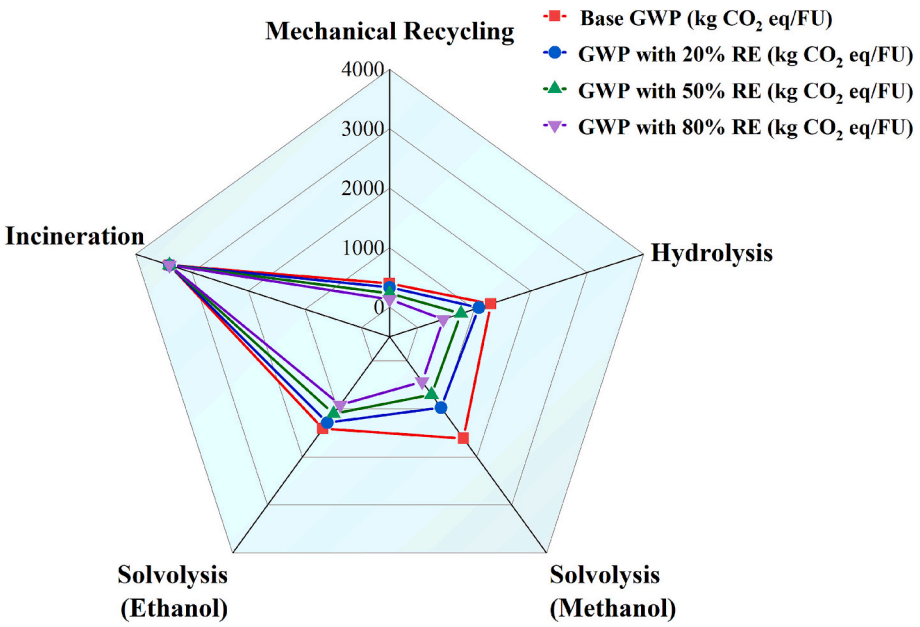


Fig. 9. Effect of partial transition to renewable energy (20 %, 50 %, and 80 %) on global warming potential across different end−of−life (EoL) pathways for PC − PLA waste (FU = 1 t PC − PLA waste).

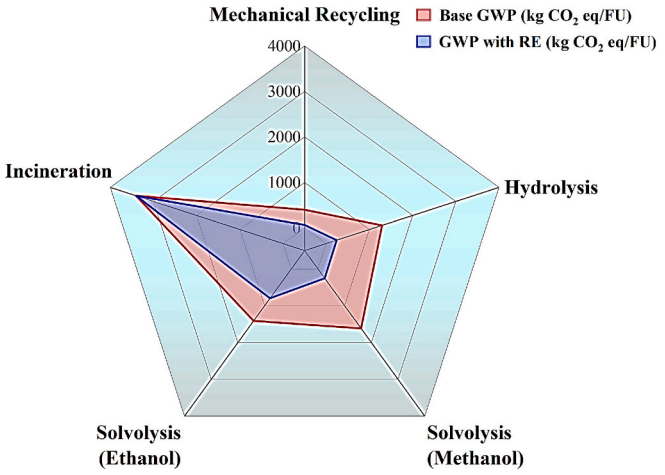


Fig. 10. Effect of 100 % transition to renewable energy on climate change in various EoL pathways for PC − PLA waste (FU = 1 t PC − PLA waste).

is provided in Table S20.

Uncertainty analysis through Monte Carlo simulation with 10,000 iterations reveals a broad range of probable impact scores across all impact categories, with no consistent pattern of variation. This highlights the inherent uncertainty in LCA results due to variability in input parameters, methodological assumptions, and data sources. For the GWP category, scenario 3 (methanolysis) exhibited the highest degree of variability, with a 95 % confidence interval ranging from 955 to 5190 kg CO_2 eq/FU (Fig. 11a). The probability histogram for scenario 3 further indicates a broad distribution of results, with a peak at 1650 kg CO_2 eq/FU (f/n = 0.0258) but extending beyond 15,000 kg CO_2 eq/FU in extreme cases (Fig. 12). These findings suggest that methanolysis not only presents substantial uncertainty in GWP estimates but also has a higher tendency to reach significantly elevated impact values compared to other scenarios. The large uncertainty range can be attributed to multiple factors, including variability in reaction conditions, catalyst efficiency, and data limitations from lab−scale studies, which introduce high uncertainty when extrapolated to industrial−scale operations. In contrast, scenario 1 (mechanical recycling) demonstrated the lowest uncertainty in GWP, with a narrower 95 % confidence interval from 121 to 1874 kg CO_2 eq/FU. The mean, median, and calculated values for scenario 1 showed relatively smaller deviations, indicating more stable results. Furthermore, its frequency histogram exhibits a higher peak than those of other scenarios, with f/n = 0.0931 at 220 kg CO_2 eq/FU (Fig. 11). The lower uncertainty in mechanical recycling is mainly due to the fewer processing steps involved, the reliance on well−established industrial data, and minimal ariability compared to chemical recycling methods.

For other impact categories, uncertainty propagation remains substantial and varies significantly across scenarios. In the AP category, scenario 4 (ethanolysis) showed the highest uncertainty, with impact scores ranging from 3 to 21 kg SO_2 eq/FU (Fig. 11b). This variability is largely influenced by differences in ethanol feedstock sources, sulfuric acid usage, catalyst regeneration efficiency, and associated SO_2 and NO_x emissions. The variability in sulfuric acid use, in particular, contributes to fluctuations in AP values, as it directly affects both emissions and the extent of acidification impacts. Similarly, for FET and CHT, chemical recycling scenarios exhibited notably higher uncertainty than mechanical recycling (Fig. 11c and d). This increased variability is primarily due to the influence of multiple factors, including electricity consumption, catalyst toxicity, and the potential leaching of unreacted chemicals or byproducts into the environment. Additionally, the sourcing of electricity from different energy mixes introduces uncertainty in toxicity−related impact categories, as fossil fuel−based power generation contributes to emissions that significantly affect ecotoxicity and human health.

The lack of objectivity and limited statistical analysis in LCA, particularly for processes highly susceptible to operational fluctuations, has been a significant concern and is often criticized by opponents of chemical recycling [53]. These concerns are further validated by the findings from our Monte Carlo simulations, which revealed substantial variability across all impact categories. This highlights the challenge of accurately assessing environmental impacts when input parameters and process efficiencies fluctuate. Among the studied scenarios, scenario 1 (mechanical recycling) exhibited the lowest uncertainty in the GWP category compared to other recycling methods. However, in absolute terms, its 95 % confidence interval remains broad, indicating considerable variability in potential outcomes. This is particularly evident

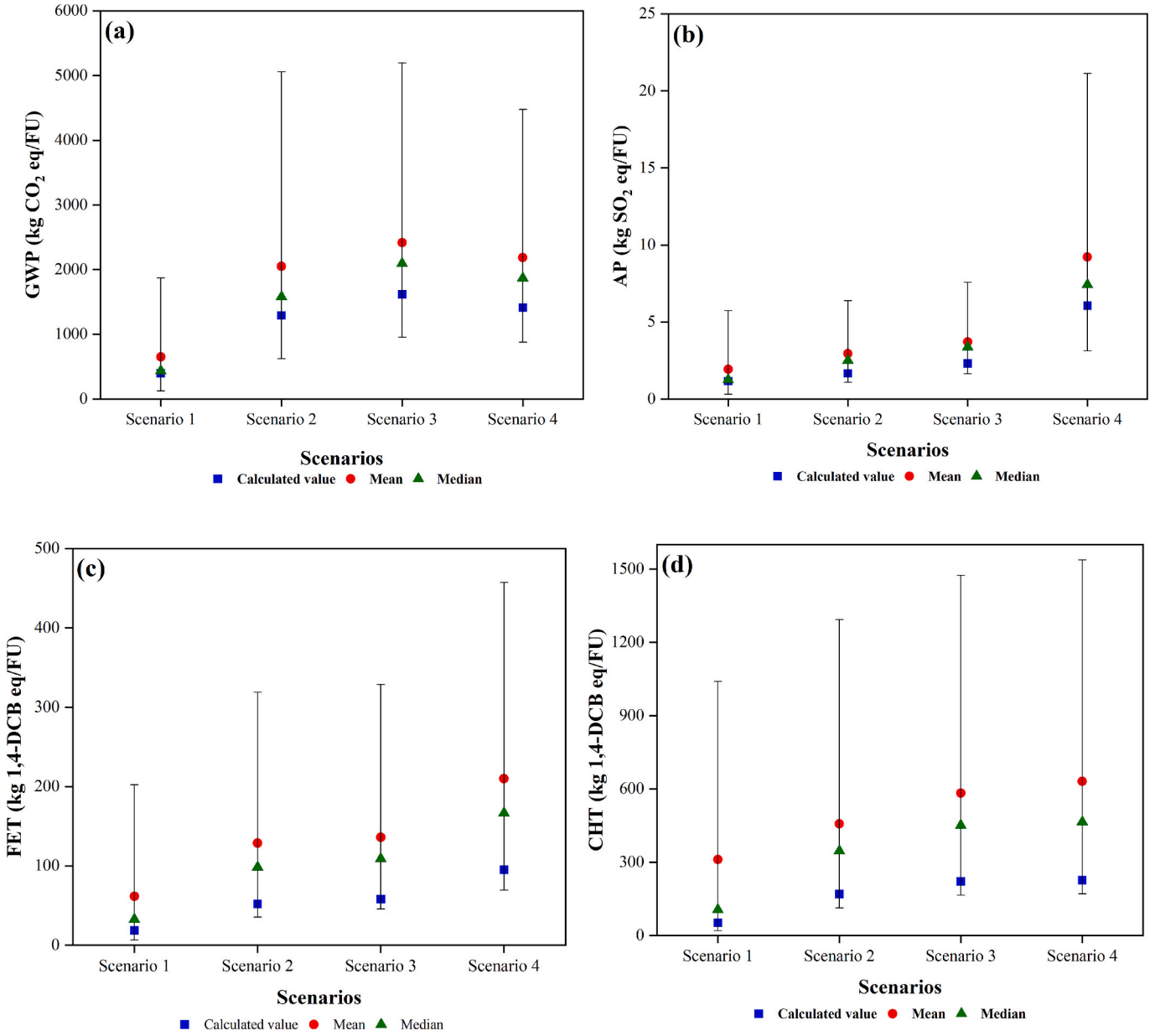


Fig. 11. Monte Carlo simulation results for four key impact categories, (a) climate change (b) acidification potential (c) freshwater ecotoxicity (d) carcinogenic human toxicity, showing the initially calculated LCA results (i.e., calculated values) as well as the propagated means and medians. The lower and upper error bars represent the 5th and 95th percentile of the result distribution, respectively. Calculation was conducted using 10,000 iterations, based on a functional unit of 1 t PC – PLA waste treated.

when comparing mechanical recycling with well–established plastic waste management methods such as incineration. Incineration demonstrated a more uniform distribution of GWP values (Fig. S5a) and had the narrowest 95 % confidence interval among all scenarios (Fig. S5b), despite its high absolute emissions. This suggests that, while incineration remains a high–impact process, its emissions are more predictable due to standardized operational conditions and established industrial–scale implementation. It is important to emphasize that this study does not advocate for incineration or open burning as a preferred waste management approach. Instead, the comparison aims to highlight differences in uncertainty levels between emerging and conventional plastic waste management methods. The higher uncertainty in chemical recycling scenarios stems from variations in reaction efficiencies, feedstock purity, and energy consumption, which are less standardized compared to incineration. Future research should focus on improving process data reliability and refining statistical methods in LCA to better capture uncertainty in novel recycling technologies.

3.5. *Case studies of PLA recycling in practice*

Several industrial case studies illustrate the real–world feasibility of mechanical and chemical recycling of PLA, aligning with the environmental priorities highlighted in this LCA. NatureWorks (USA) has implemented both mechanical sorting and chemical hydrolysis at its Blair facility, recovering over 17 million pounds of PLA back into lactic acid for re – polymerization, effectively enabling closed–loop recycling with near–complete monomer recovery [54]. However, the company's early attempt to commercialize PLA water bottles was curtailed by end–of–life obstacles. In 2007, NatureWorks placed a hold on expanding its Ingeo™ PLA bottles after US recyclers and NGOs requested a moratorium, warning that PLA containers could contaminate PET recycling streams due to inadequate sorting technology [54]. This

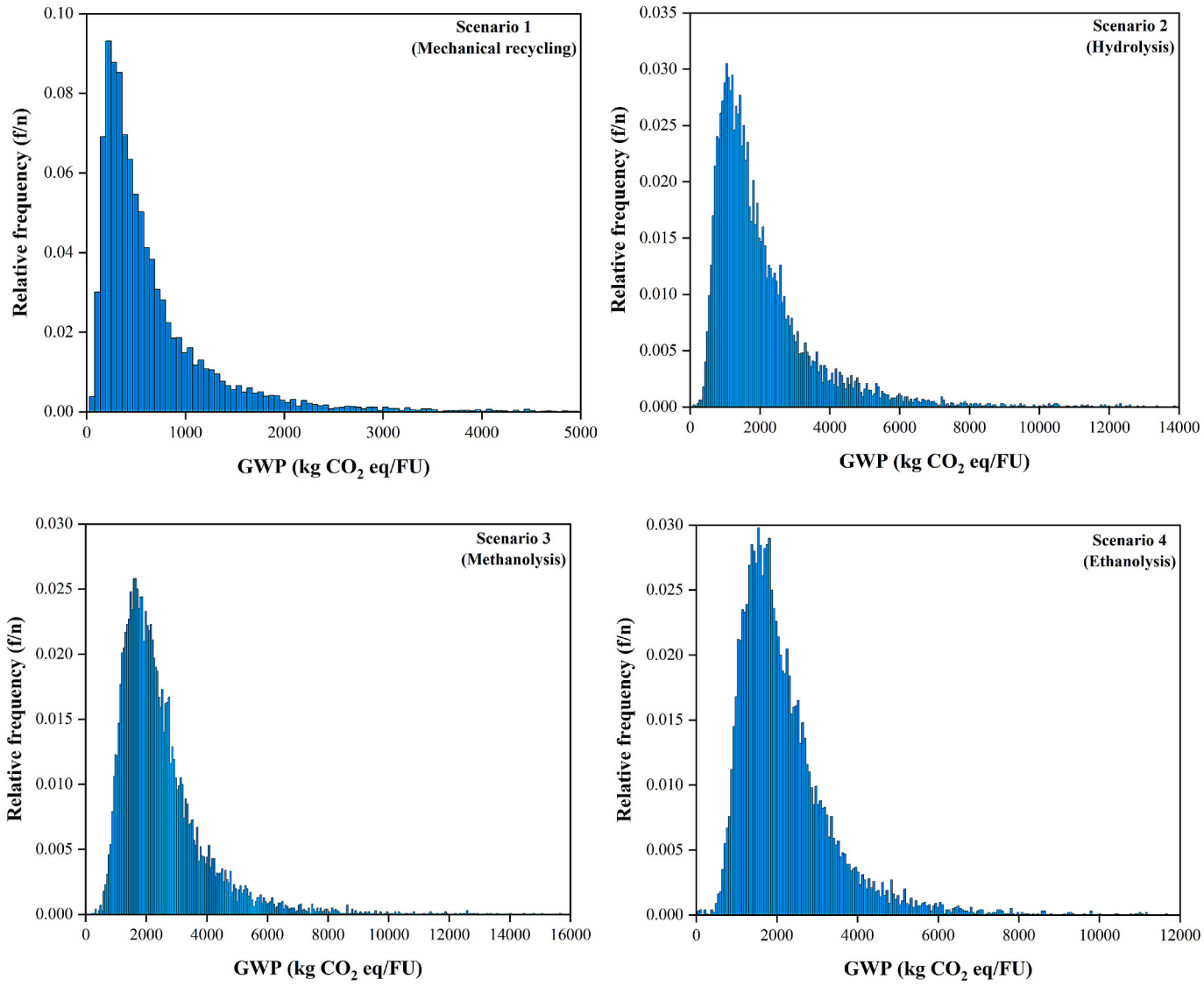


Fig. 12. Probability histograms for global warming potential (unit: kg CO_2 eq/FU; bin size: 55 kg CO_2 eq/FU). All distributions were right−skewed, indicating that while most GWP values are concentrated in lower ranges, each scenario has potential for much higher GWP outcomes.

pauses in bottle roll−out illustrates how technical limitations in waste sorting and industry pushback forced NatureWorks to shelve a PLA bottle−to−bottle recycling initiative.

Similarly, Total Corbion has developed Luminy® rPLA using a depolymerization−based process that converts post−consumer PLA bottles into virgin−equivalent resin, demonstrating a bottle−to−bottle model with reduced life−cycle emissions. While promising, both initiatives face scalability limitations due to dependence on clean waste streams and limited infrastructure [55]. Nonetheless, The Total–Corbion joint venture recently faced a major setback in its PLA expansion plans. In mid − 2023, the venture cancelled a proposed new PLA production and recycling plant in Grandpuits, France, citing surging construction costs and uncertainty in the market climate [55]. This withdrawal of a high−profile project underscores how economic and market factors can derail the scale−up of PLA recycling infrastructure even for industry leaders.

Complementing these, Carbios (France) has pioneered enzymatic depolymerization of PLA, achieving ~90 % monomer recovery within 48 h under mild conditions, including from semi−crystalline waste. This approach is particularly suited for mixed or contaminated streams, offering high−purity outputs with minimal thermal degradation [56]. However, the French company Carbios, a pioneer of enzymatic polymer recycling, has experienced delays in translating its PLA recycling innovations to full−scale deployment. The company demonstrated efficient enzyme−catalyzed depolymerization of PLA at lab and pilot scale [56], and even introduced an enzyme additive (Evanesto®) in 2020 to make PLA − based plastics compostable in home composting conditions [57]. Nevertheless, Carbios's PLA − focused processes remain at a pre − commercial stage, notably, its first industrial bio − recycling plant under construction is dedicated to PET, not PLA. This reflects ongoing scale−up challenges and a strategic prioritization of larger polymer markets over PLA recycling efforts.

3.6. *Implementation roadmap for recycling strategy selection*

To enable the practical deployment of optimal PLA recycling strategies, a comprehensive implementation framework is proposed (Fig. 13), integrating waste−specific characteristics, infrastructure readiness, and environmental performance metrics. The approach begins with assessing the local PLA waste stream, including the quantification of generation rates, contamination levels, and sorting capabilities, which directly influence the feasibility and efficiency of downstream processing pathways. This is followed by evaluating regional infrastructure and energy profiles to determine technological

compatibility and operational limitations. Key parameters include the availability and processing capacity of recycling facilities, transportation logistics, the composition of the regional energy mix (e.g., proportion of renewables), and the technical preparedness of the workforce to operate mechanical or chemical recycling systems.

To support data−driven decision−making, the roadmap incorporates energy efficiency standards based on Specific Energy Consumption (SEC), expressed in megajoules per kilogram of PLA processed (MJ/kg). Mechanical recycling, with a typical SEC of 0.8–2 MJ/kg, is highly energy−efficient and generally preferred in regions with limited renewable energy access. In contrast, chemical recycling methods such as hydrolysis and alcoholysis demand significantly higher energy inputs, ranging from 12 to 25 MJ/kg, due to thermal reaction conditions and solvent recovery requirements. As such, chemical recycling becomes more viable in areas where the local grid has a high share of renewable energy (e.g., >60 %) or where high−purity rPLA output is essential. To operationalize this criterion, threshold−based guidance is recommended: if SEC exceeds 10 MJ/kg and the regional energy mix contains less than 40 % renewables, mechanical recycling should be prioritized. Conversely, chemical recycling may be favoured in low−carbon energy contexts, particularly when solvent recovery efficiency reaches ≥90 %. This energy benchmarking enhances the environmental credibility of the roadmap and aligns recycling decisions with broader decarbonization goals.

Technology selection is further guided by LCA outcomes and contamination thresholds. Mechanical recycling is prioritized for contamination levels ≤2–3 %, while chemical recycling is more appropriate for moderately contaminated streams (3–10 %). Streams exceeding 10 % contamination are generally excluded from material recovery and may be directed to thermal valorisation. Additionally, local policy instruments and market dynamics for rPLA influence strategic choices. Successful implementation requires robust quality management protocols, including defined performance parameters for rPLA (e.g., melt flow index, molecular weight), solvent recovery targets (≥90 %), and compliance with relevant regulatory standards (e.g., FDA, EU food−contact approvals). Finally, ongoing monitoring of performance indicators, such as material recovery rate, environmental impacts (e.g., GWP), rejection rates, and process losses, facilitates adaptive process optimization. This feedback−driven approach ensures long−term operational sustainability and enables alignment with evolving technical, regulatory, and market landscapes.

4. Limitation of this study

This study has several limitations related to data availability, modelling assumptions, methodological choices, and system−level constraints, which should be considered when interpreting the results. A significant limitation stems from the reliance on laboratory−scale or literature−sourced data for modelling the chemical recycling pathways. While such data are commonly used in early−stage LCA studies, they may not accurately reflect the scalability, process efficiency, or operational complexity of industrial−scale systems. Extrapolating from lab−−scale datasets introduces uncertainty, especially when integrating data from sources with varying system boundaries, operating conditions, and reaction efficiencies. These limitations affect the accuracy of energy and material balances, which are critical for comparative assessments. The process simulation in Aspen Plus further required assumptions about conversion rates, reaction yields, and energy inputs, which, although informed by literature, may oversimplify the realities of full−scale implementation. Additionally, mechanical and chemical recycling routes depend on high−purity PLA feedstock, yet real−world sorting and collection infrastructure is often insufficient to provide uncontaminated input material, impacting both environmental performance and technical feasibility. Methodological constraints, including allocation rules and the impact assessment models used, also introduce uncertainty, particularly for emissions and toxicity−related impacts. Furthermore, toxicity impact assessments remain uncertain due to limitations in models like USEtox, which exclude occupational exposure [58] and lack emission−specific data such as solvent release. Although sensitivity and Monte Carlo analyses were used to quantify and propagate uncertainties, these techniques cannot fully compensate for the underlying limitations of incomplete, non − industrial data and generalized assumptions. Moreover, as recycling technologies mature, continuous model refinement and validation against real−world industrial data will be essential to improve result accuracy and applicability. Despite these constraints, the study provides important insights into the comparative environmental impacts of various PLA end−of−life strategies and highlights key directions for future research and technological development.

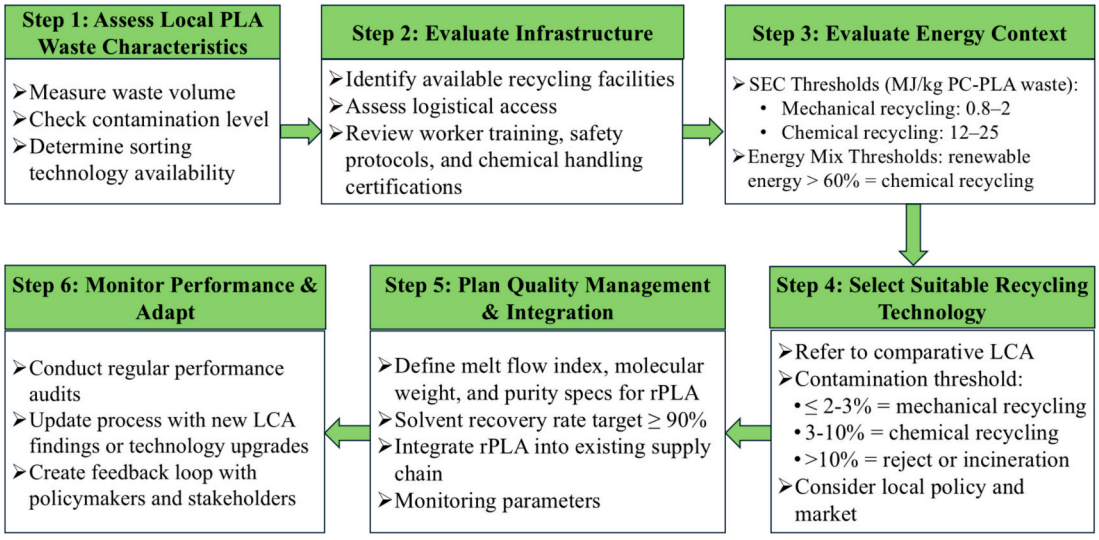


Fig. 13. Implementation roadmap for recycling strategy selection for post−consumer PLA. The roadmap outlines a five−step decision−making framework integrating LCA results with local feasibility factors. It includes: (1) assessment of PLA waste volume, contamination, and segregation efficiency; (2) evaluation of recycling infrastructure, labour availability, (3) evaluation of energy mix; (4) selection of appropriate recycling technology based on environmental performance and material quality needs; (5) planning of quality control measures and process integration; and (6) monitoring of environmental and economic outcomes with flexibility to adapt and blend with virgin PLA if required.

5. Future recommendations

To enhance the accuracy and applicability of LCA studies on PC − PLA waste recycling, future research should prioritise the collection of primary data from industrial−scale recycling operations to improve model reliability and reduce uncertainty. Strengthening PC − PLA collection and sorting infrastructure—through advanced techniques such as NIR spectroscopy and enzymatic separation—will be critical for ensuring feedstock purity and recycling efficiency. Policy measures, including extended producer responsibility (EPR) schemes and standardized labelling for biodegradable plastics, can support upstream segregation and improve recycling rates. Further optimisation of chemical recycling methods (e.g., hydrolysis, methanolysis, ethanolysis) is needed to improve energy efficiency and reduce environmental impacts. This requires large−scale pilot testing, process simulation, and techno−economic analysis. Incorporating renewable energy into recycling operations offers additional sustainability benefits.

While this study focused on recycling and thermal EoL pathways for PC − PLA, future work will also explore the environmental implications of additional scenarios such as composting and landfilling, which are commonly associated with PLA due to its compostable nature. A forthcoming manuscript will address these pathways using an LCA approach, enabling a more comprehensive understanding of EoL options for bioplastics. Future studies should also adopt dynamic LCA approaches that consider evolving technologies, regulatory landscapes, and market conditions, enabling more robust assessment of long−term EoL scenarios for bioplastics.

6. Conclusions

This LCA study demonstrates that all assessed recycling strategies for post−consumer PLA, mechanical, hydrolytic, and alcoholytic, consistently outperform incineration across major environmental impact categories, including global warming, acidification, eutrophication, and land use. Mechanical recycling exhibits the lowest overall environmental burden due to its limited energy consumption and low emissions, making it an effective and accessible solution where clean PLA waste streams are available. However, its long−term applicability is constrained by polymer degradation through repeated cycles. In contrast, chemical recycling approaches, such as hydrolysis and alcoholysis, offer greater potential for high−quality monomer recovery and closed loop material reuse, despite their higher energy requirements and current technological limitations. While incineration provides some energy recovery benefits, it incurs significantly greater environmental penalties and should only be considered when recycling options are not viable. Based on these findings, mechanical recycling is recommended as the most practical and sustainable option for small−to−medium enterprises and localized waste management systems. For industrial−scale operations, particularly those aiming for high material purity and long−term circularity, chemical recycling pathways offer a more suitable route, provided that energy inputs can be optimized and system efficiencies improved. These results provide a robust foundation for guiding both policy and infrastructure development and highlight the need for tailored waste management strategies based on scale, feedstock quality, and resource availability. Beyond immediate application, this study also establishes a methodological baseline and comparative reference that future research can build upon. Furthermore, the sensitivity and Monte Carlo uncertainty analyses underscore the influence of process parameters, energy sources, and data quality on the environmental outcomes, reinforcing the importance of high−resolution, context−specific data in future assessments. Overall, this study supports the advancement of PLA recycling technologies and suggests clear research priorities in energy integration, process improvement, and commercial−scale validation to inform more effective and sustainable waste management practices.

CRediT authorship contribution statement

Md. Monjurul Islam: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Nawshad Haque:** Writing – review & editing, Supervision, Funding acquisition. **Deborah Lau:** Writing – review & editing, Supervision, Funding acquisition. **Muhammed Bhuiyan:** Writing – review & editing, Supervision. **Biplob Kumar Pramanik:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of Generative AI and AI−assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT to assist with paraphrasing, improving language clarity, and enhancing overall readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2025.167057.

Data availability

All the data used and generated can be found in the text and Supplementary information.

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